

**Introduction to
Information &
Communication
Technologies
CL-1000**

Lab 01

**Features of MS Word and MS
Excel**

National University of Computer & Emerging Sciences – NUCES – Karachi



National University of Computer & Emerging Sciences – NUCES – Karachi

Course Code: CL-1000

Introduction to Information & Communication
Technologies

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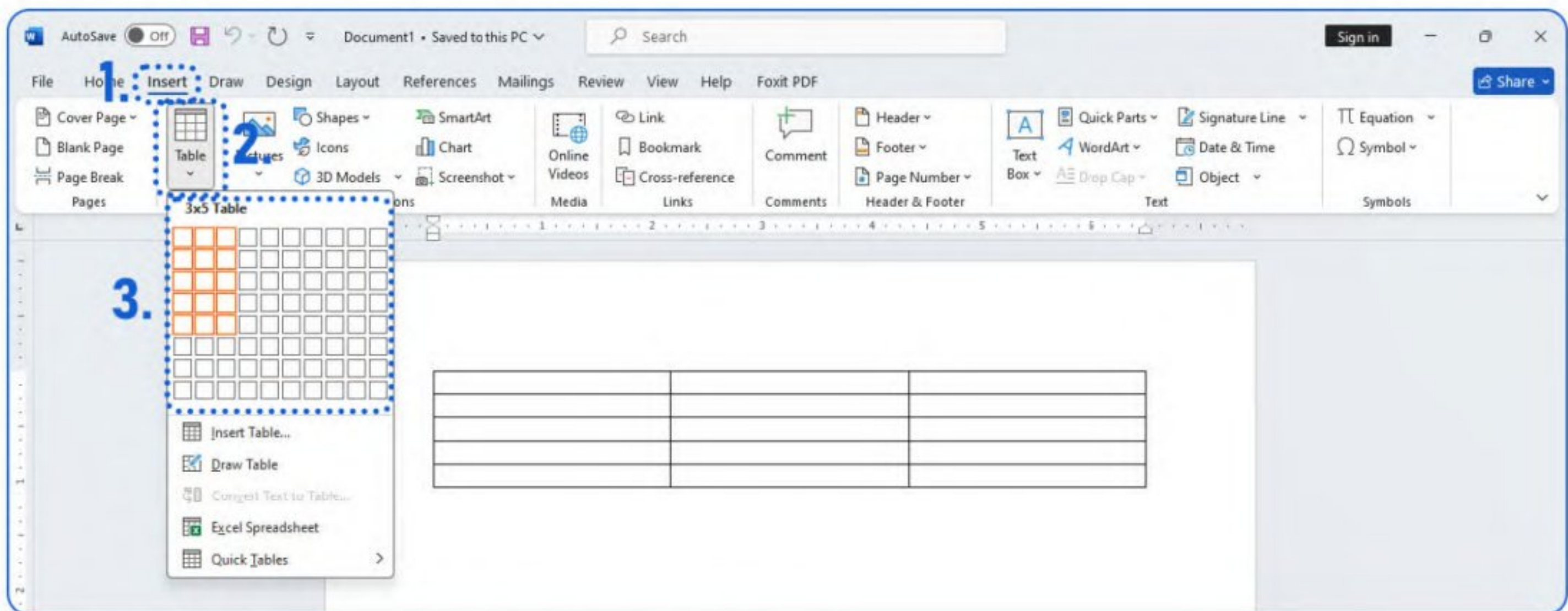
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1. Table Operations

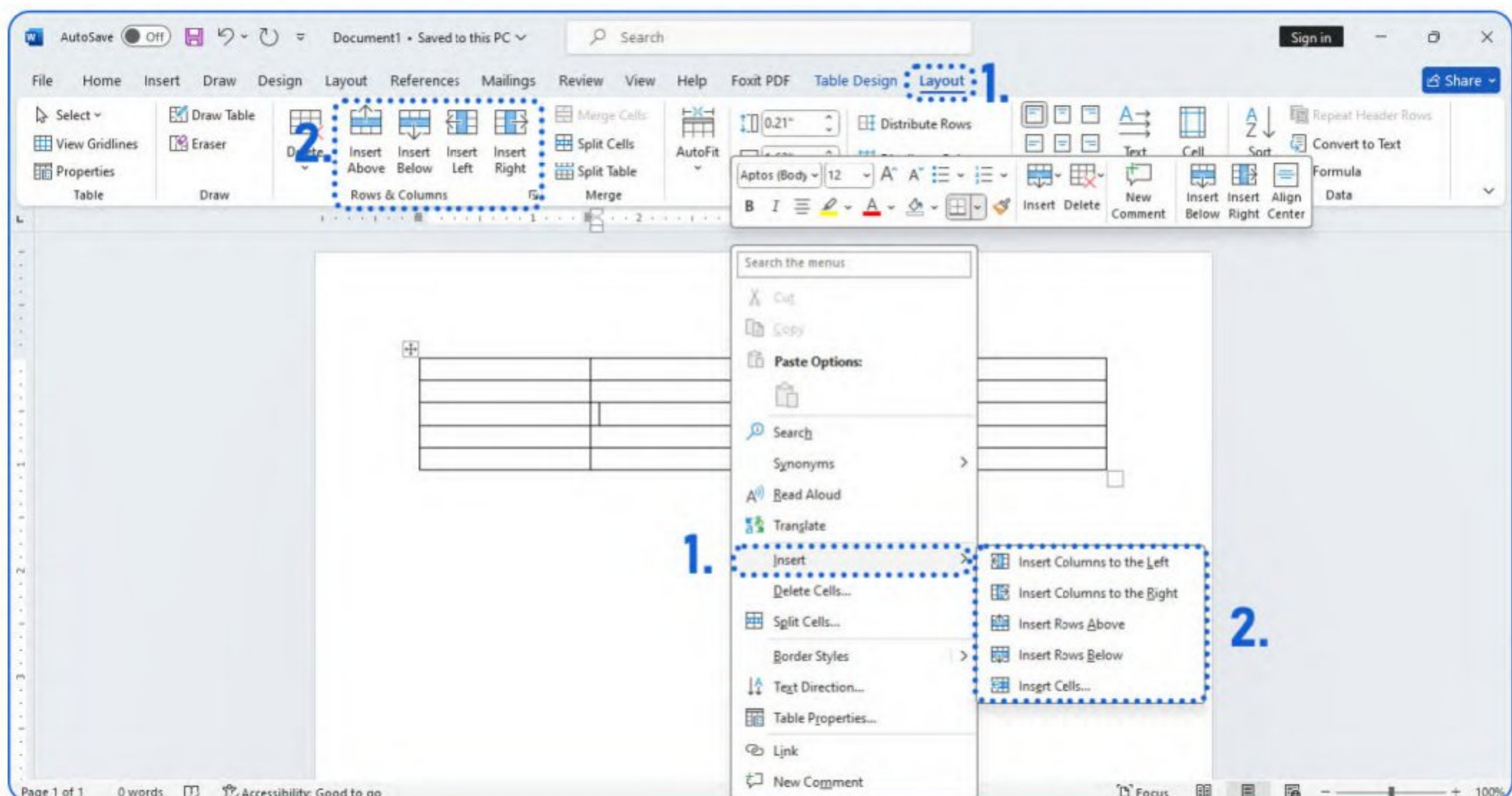
1.1 Creating a Table

1. Click on the **Insert** tab.
2. Click on **Table**.
3. Select the grid size by hovering the cursor on the grid. The desired table will be inserted.
4. You can also manually define the number of rows and columns of a table by click on **Insert Table...**



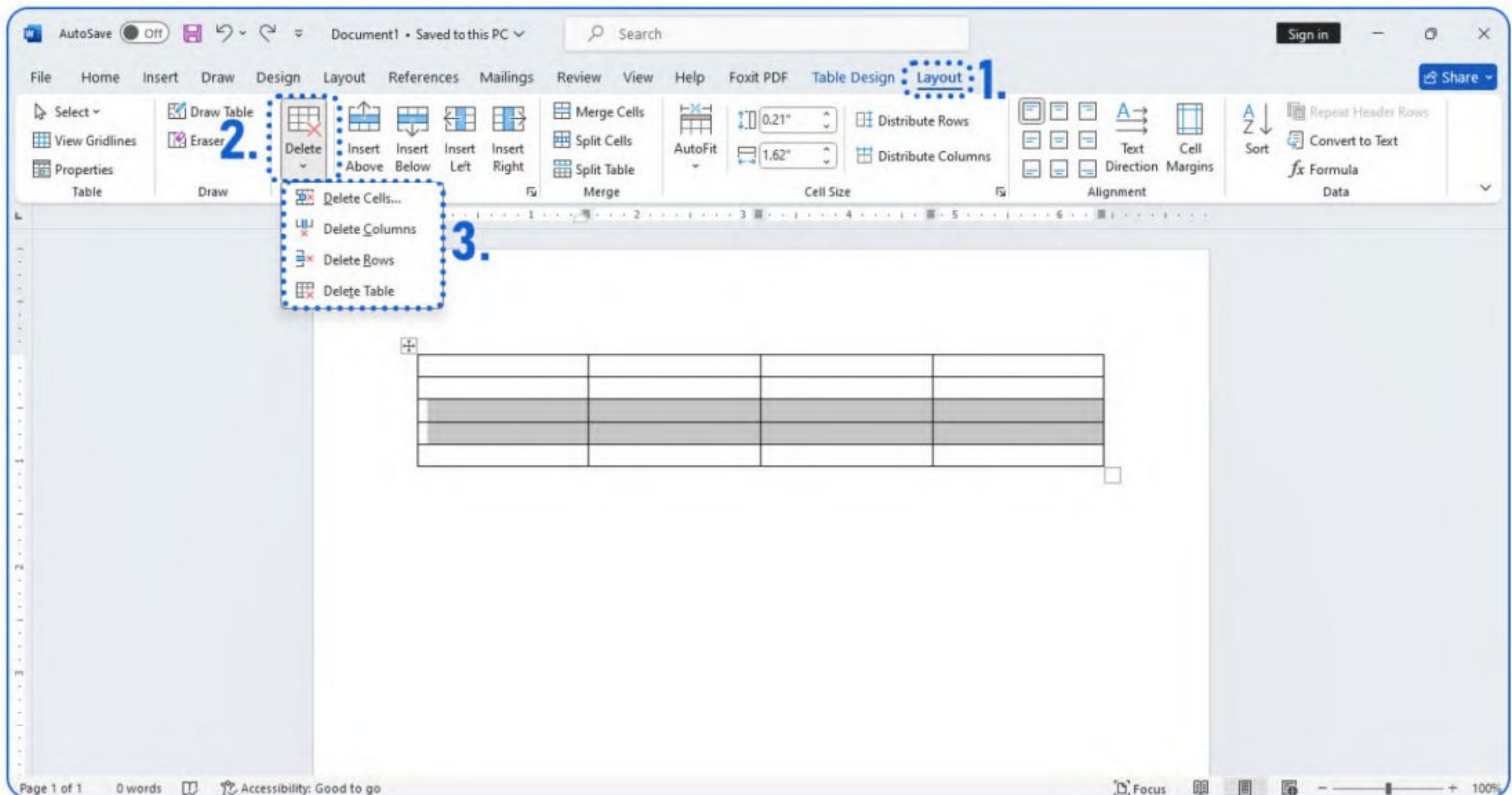
1.2 Inserting Rows and Columns in a Table

1. After inserting the table, a tab named **Layout** will appear. Click on it.
2. You can insert the rows and columns at the desired direction from the **Rows & Columns** section. Or you can right-click to a **row/column/cell**, then click on **insert** to do the same.



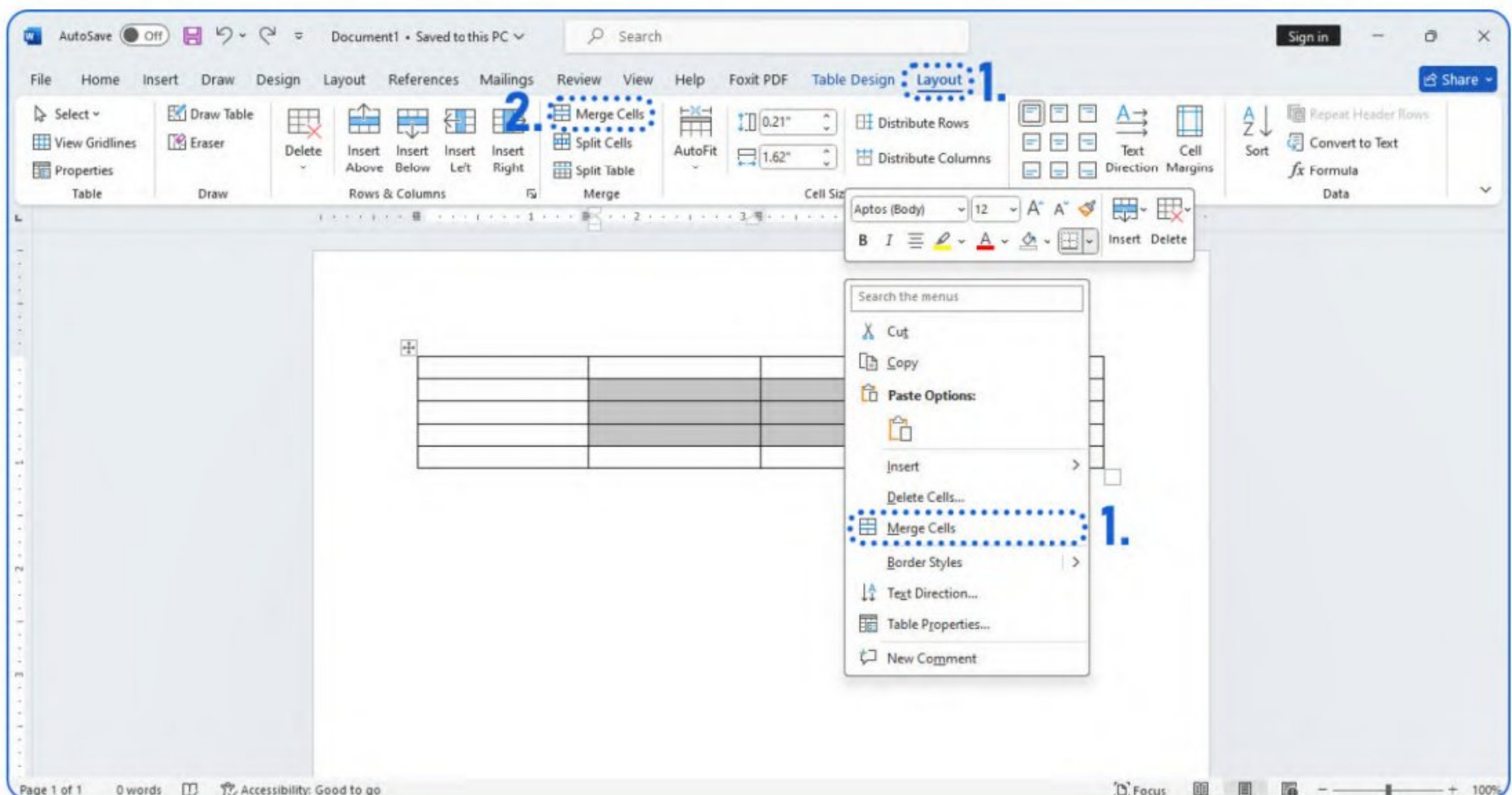
1.3 Deleting Rows and Columns of a Table

1. Select the desired row(s)/columns(s)/cells you want to delete.
2. Under the **Layout** tab, click on **Delete**. A drop-down menu will appear which allows you to delete the cells, rows, columns or the entire table.



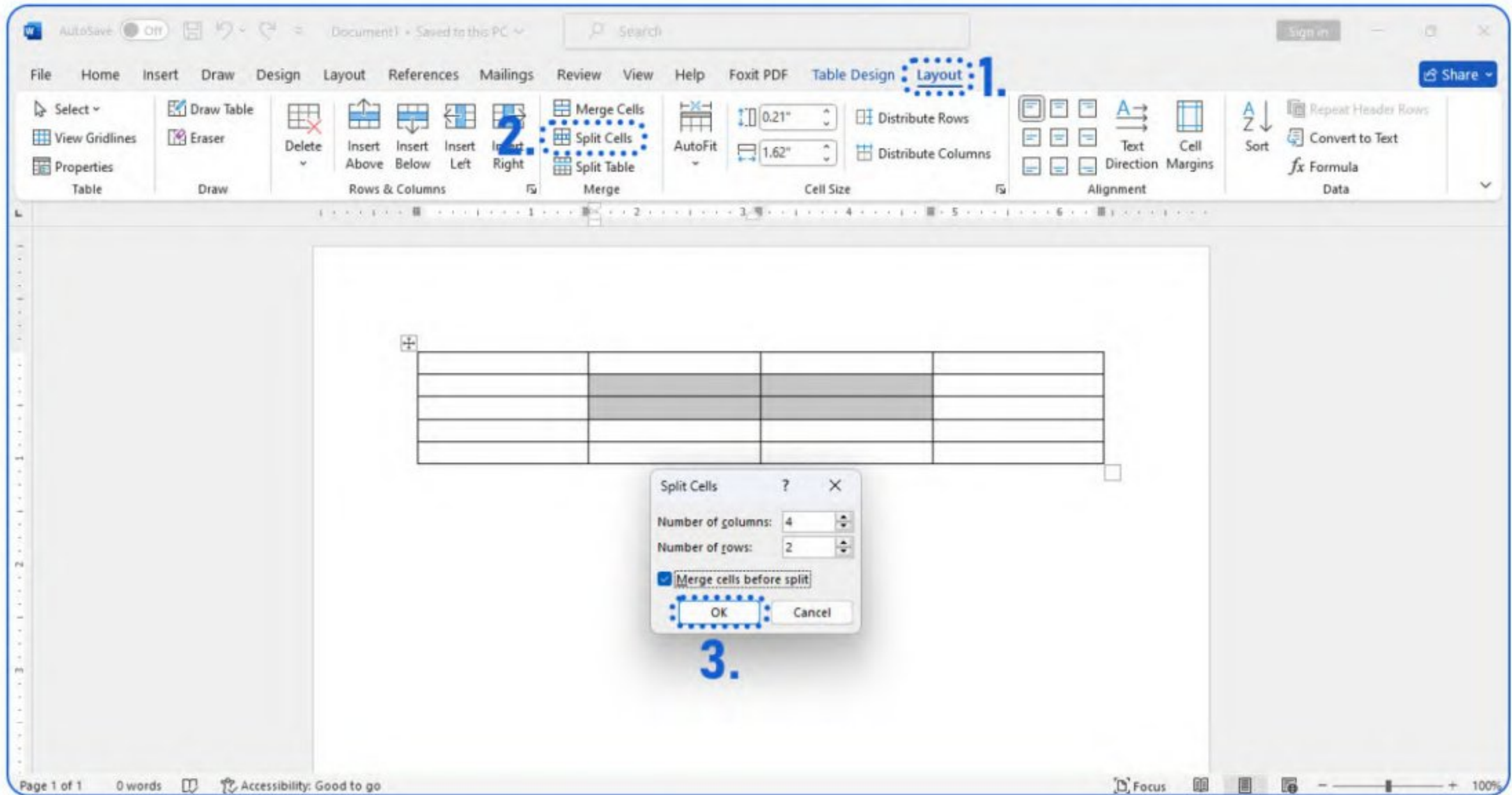
1.4 Merging Cells

1. Select the cells you want to merge.
2. Under the **Layout** tab, click **Merge**. Or, you can right-click on the selected cell and click on **Merge Cells**.



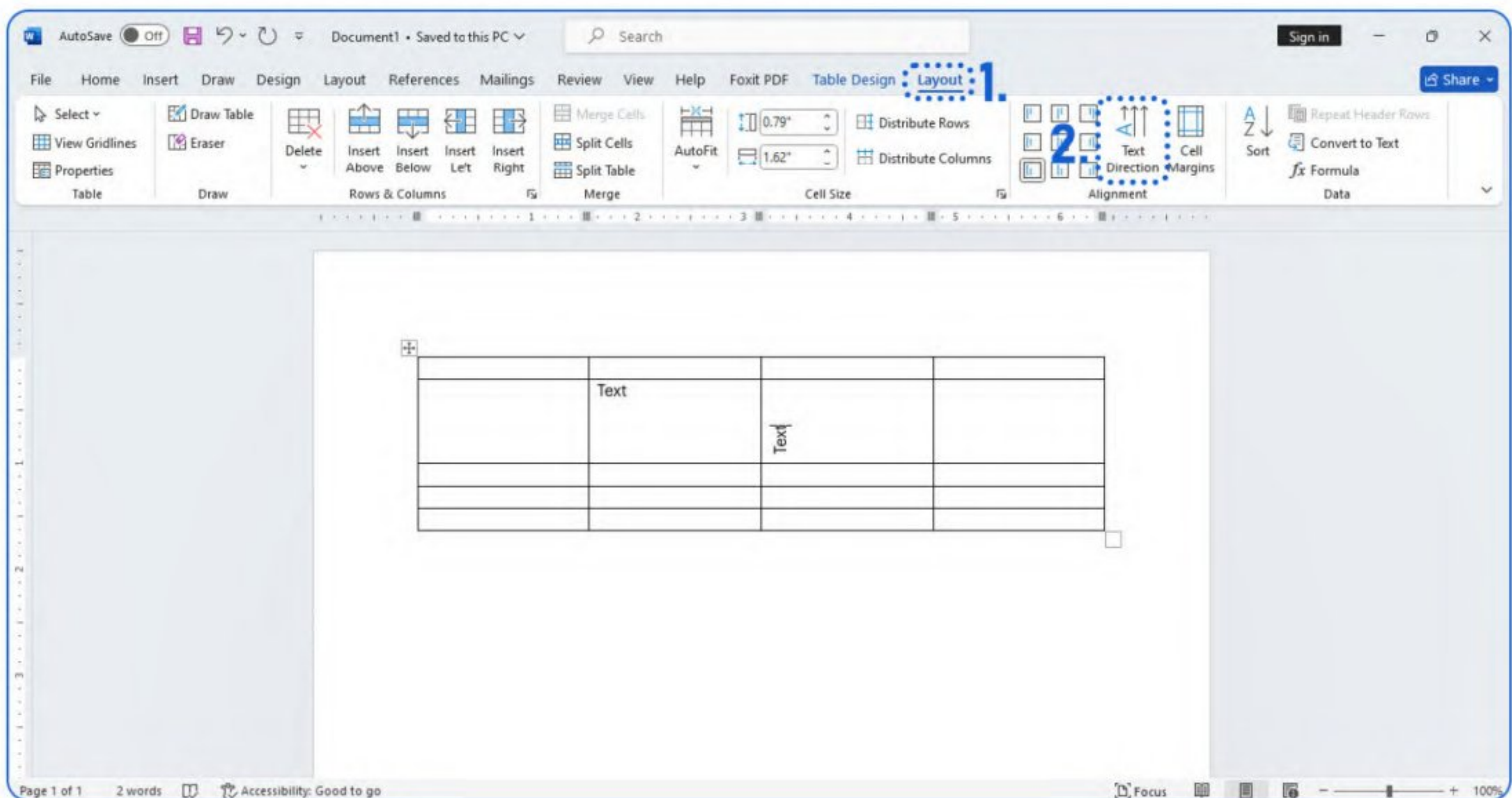
1.5 Splitting Cells

1. Select the cells you want to split.
2. Under the **Layout** tab, click on **Split**. A window will appear.
3. Insert the number of rows and columns in which you want to split the cells.



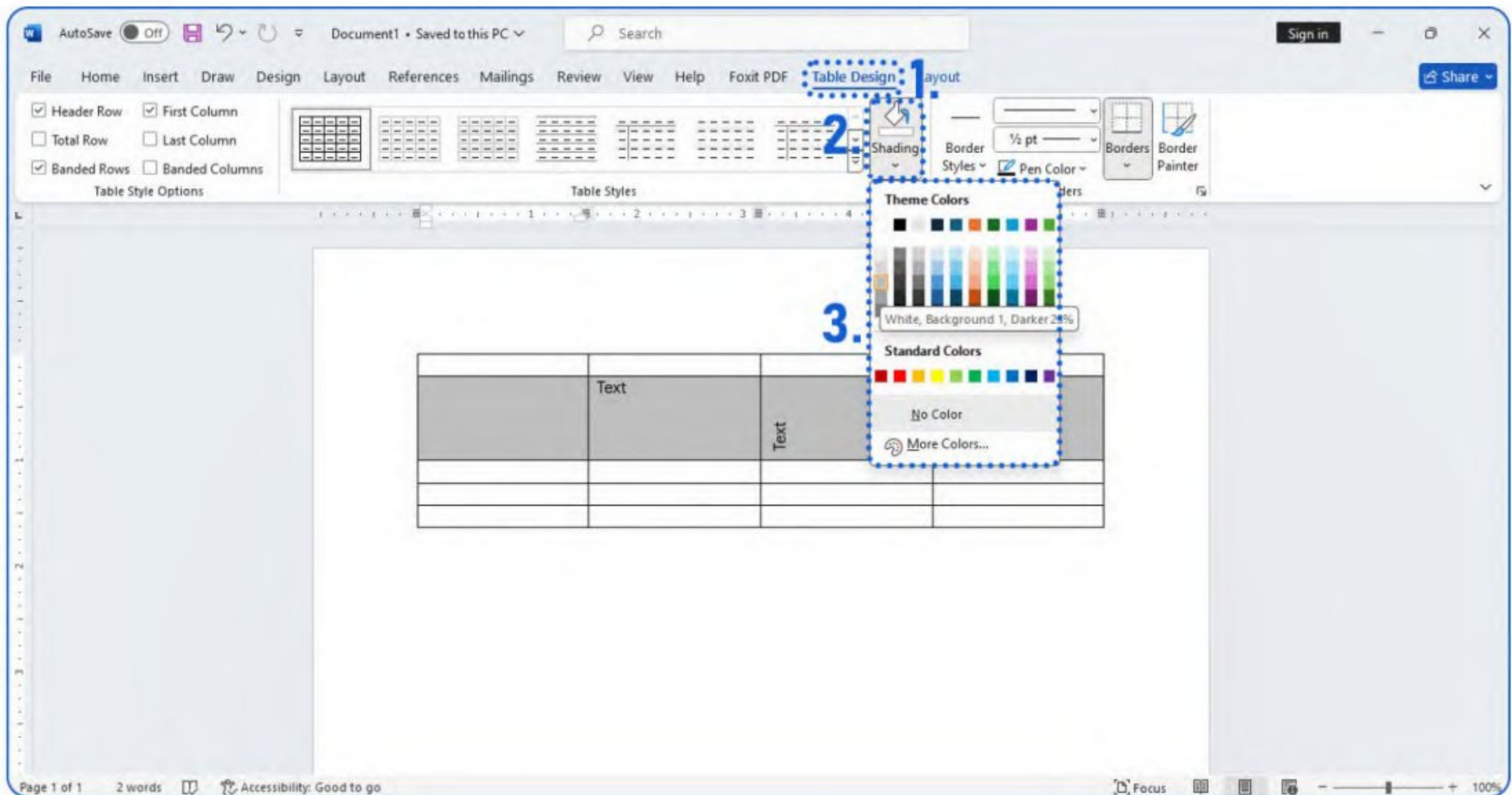
1.5 Changing Direction of Text in a Cell

1. Select the cell(s) to which you want to change the text direction.
2. Under the **Layout** tab, in the **Alignment** section, click on **Text Direction** to change the direction of the text to the direction you desire.



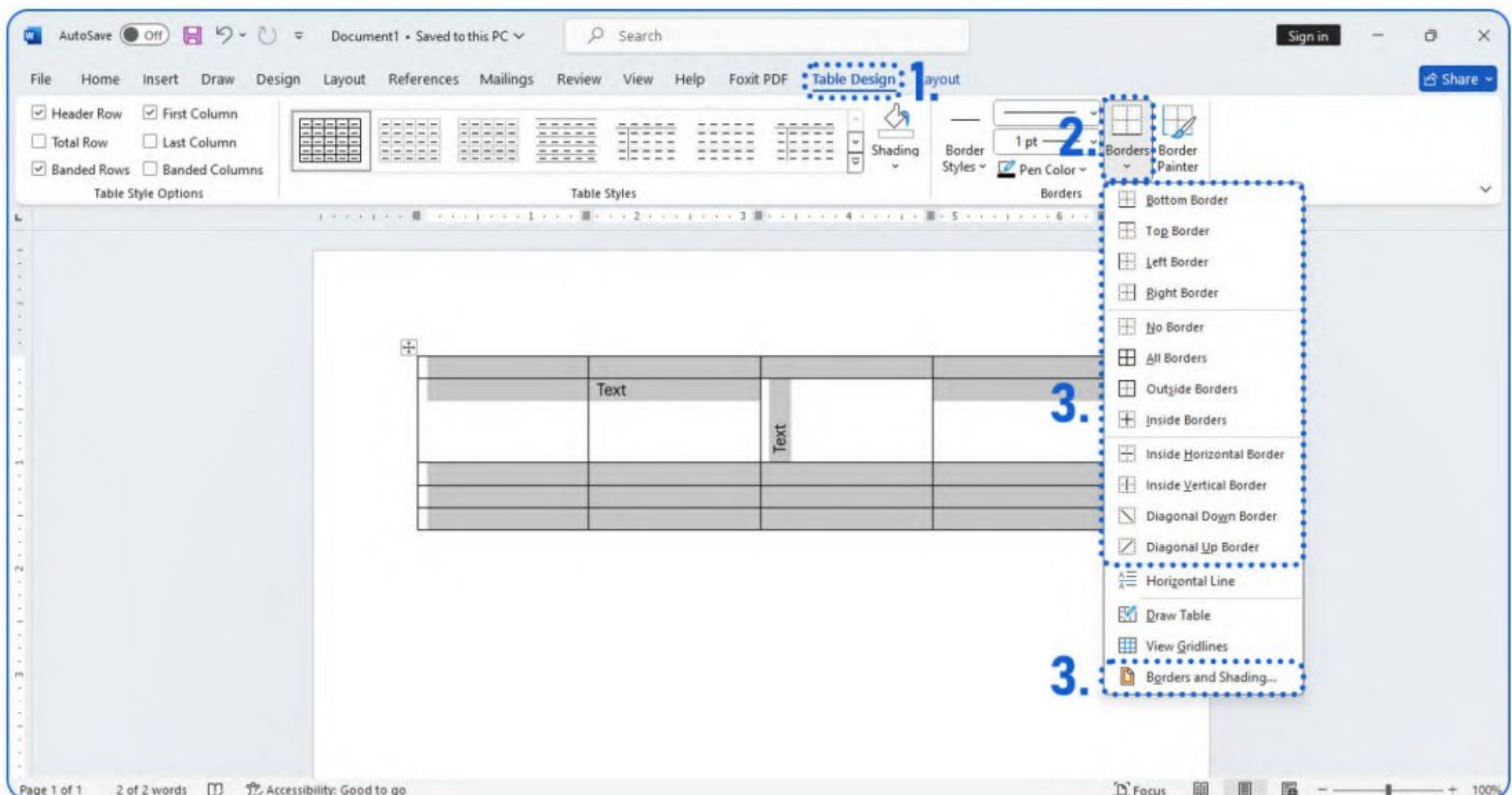
1.6 Apply Shading to Cells, Rows, and Columns

1. Select the row(s), columns(s) or cell(s) to apply shading on them.
2. Click on **Table Design** tab.
3. Click on **Shading** and choose your desire colour.

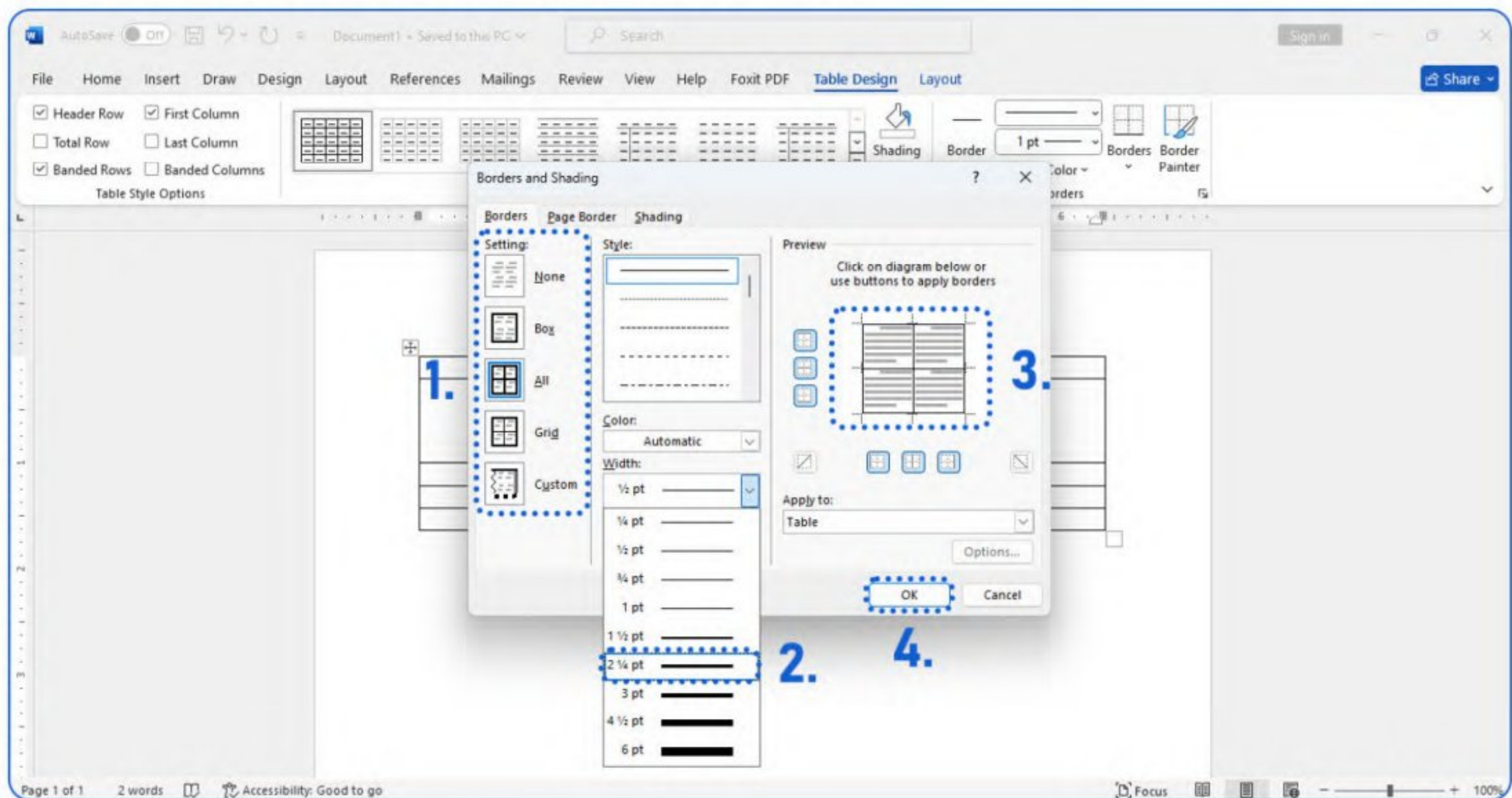


1.7 Apply Borders to Cells, Rows, and Columns

1. Select the row(s), columns(s) or cell(s) to apply border on them.
2. Under the **Table Design** tab, click on **Borders**.
3. Click on **Border and Shading...**

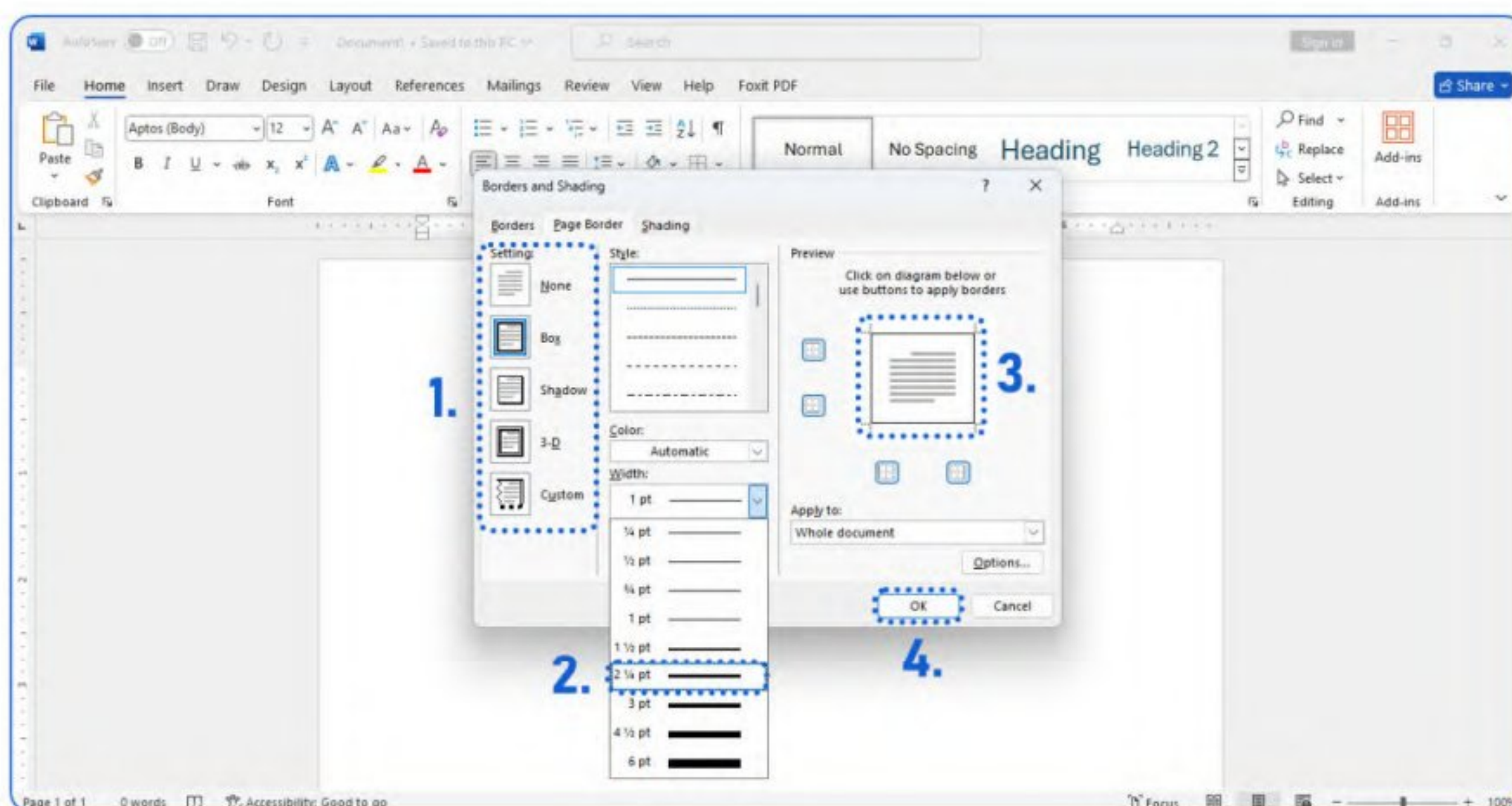


4. The **Border and Shading** window will open.
5. You can choose the border preset from **Setting** label, border style from the **Style** label, thickness of the border from the **Width** label and sides to which you want to apply the border.
6. Click on **OK** after the changes you have made.



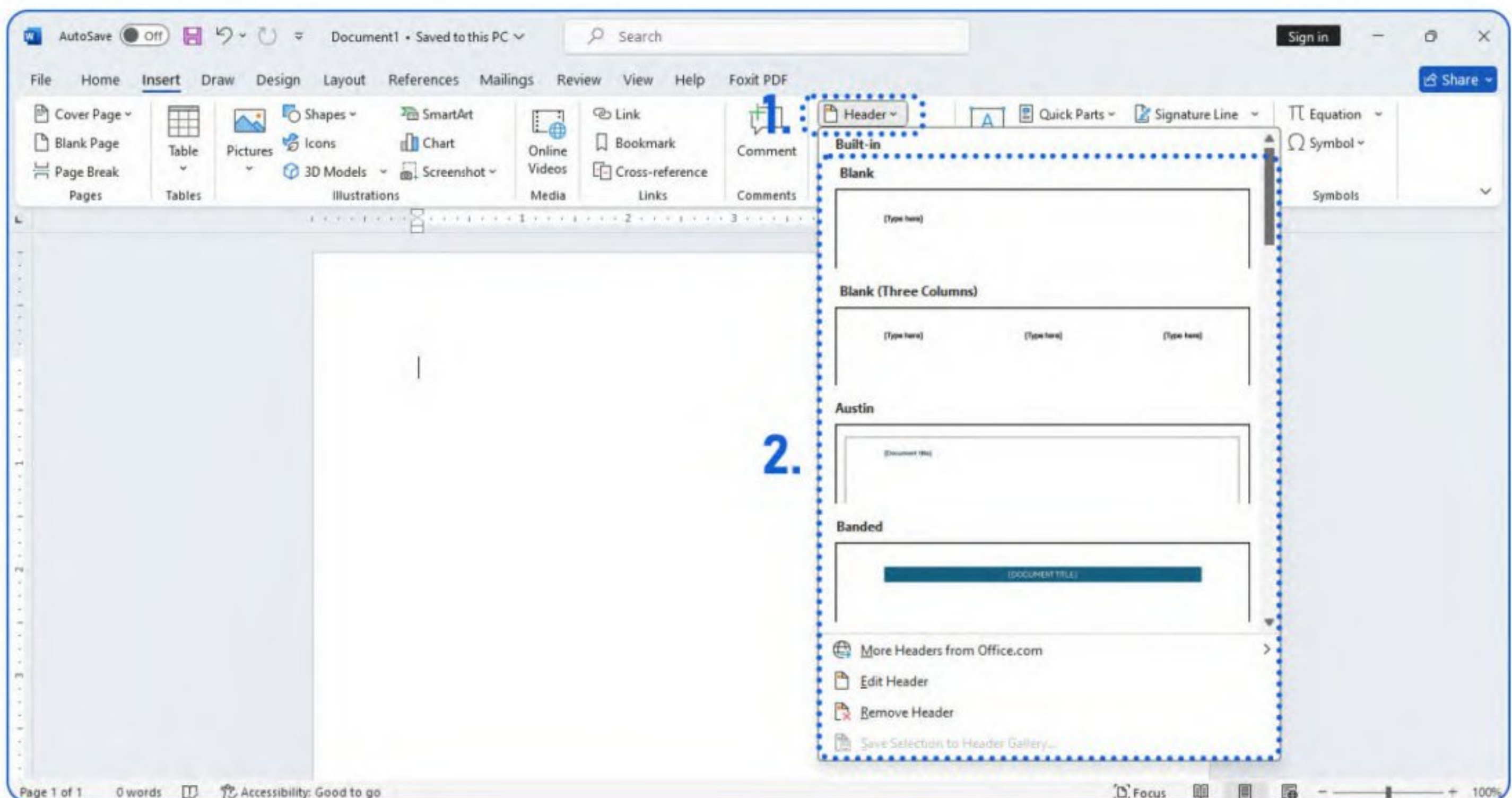
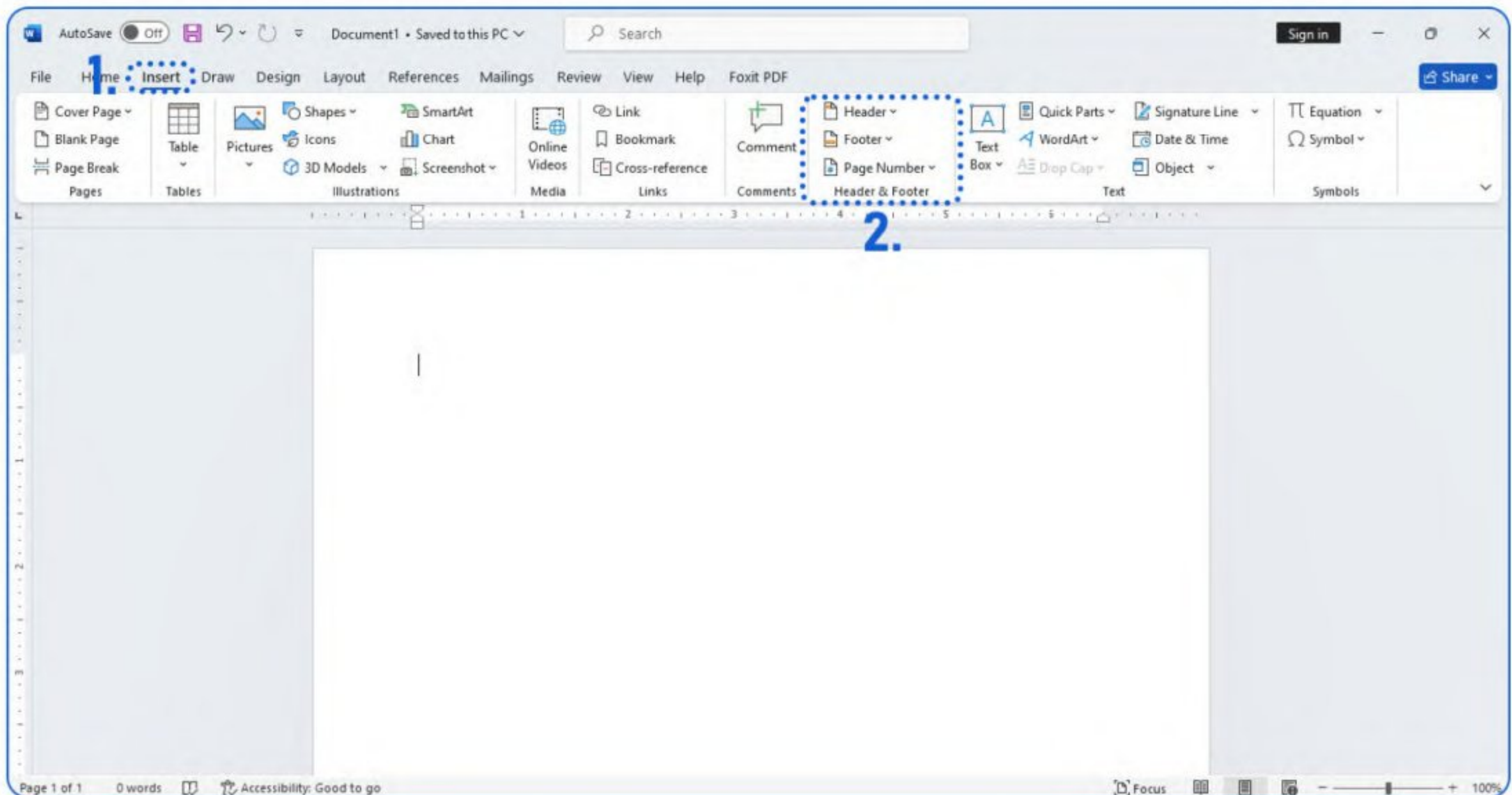
2. Apply Borders to a Page

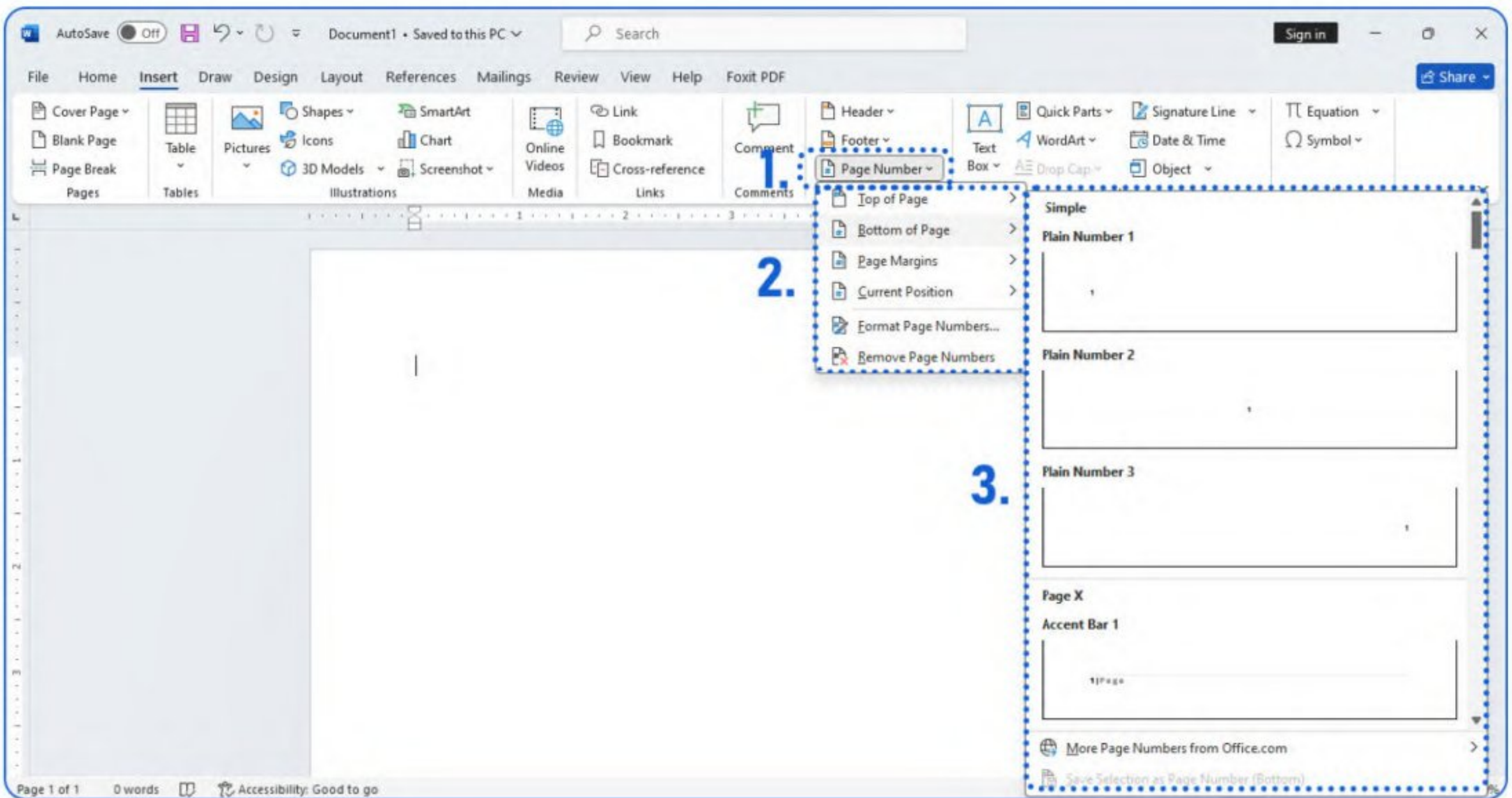
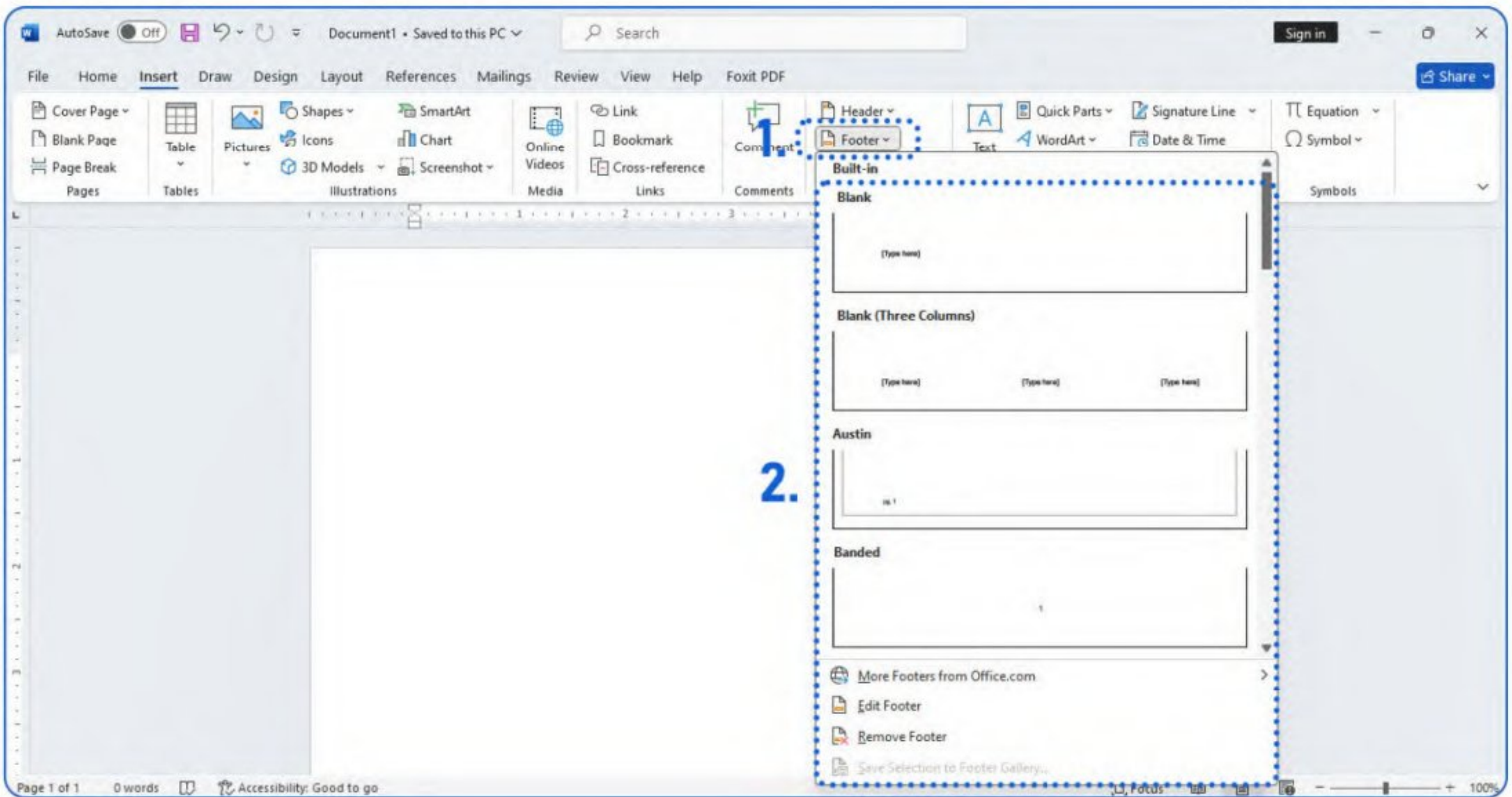
1. Click on the **Home** tab.
2. In the **Paragraph** section, click on the **dotted-box**, located in the second row at right-most side of the section.
3. The **Borders and Shading** window will appear. Click on the **Page Border** tab.
4. Follow the same steps as you have followed to apply borders on tables. (see section 1.7)



3. Apply Header, Footer and Page Numbers

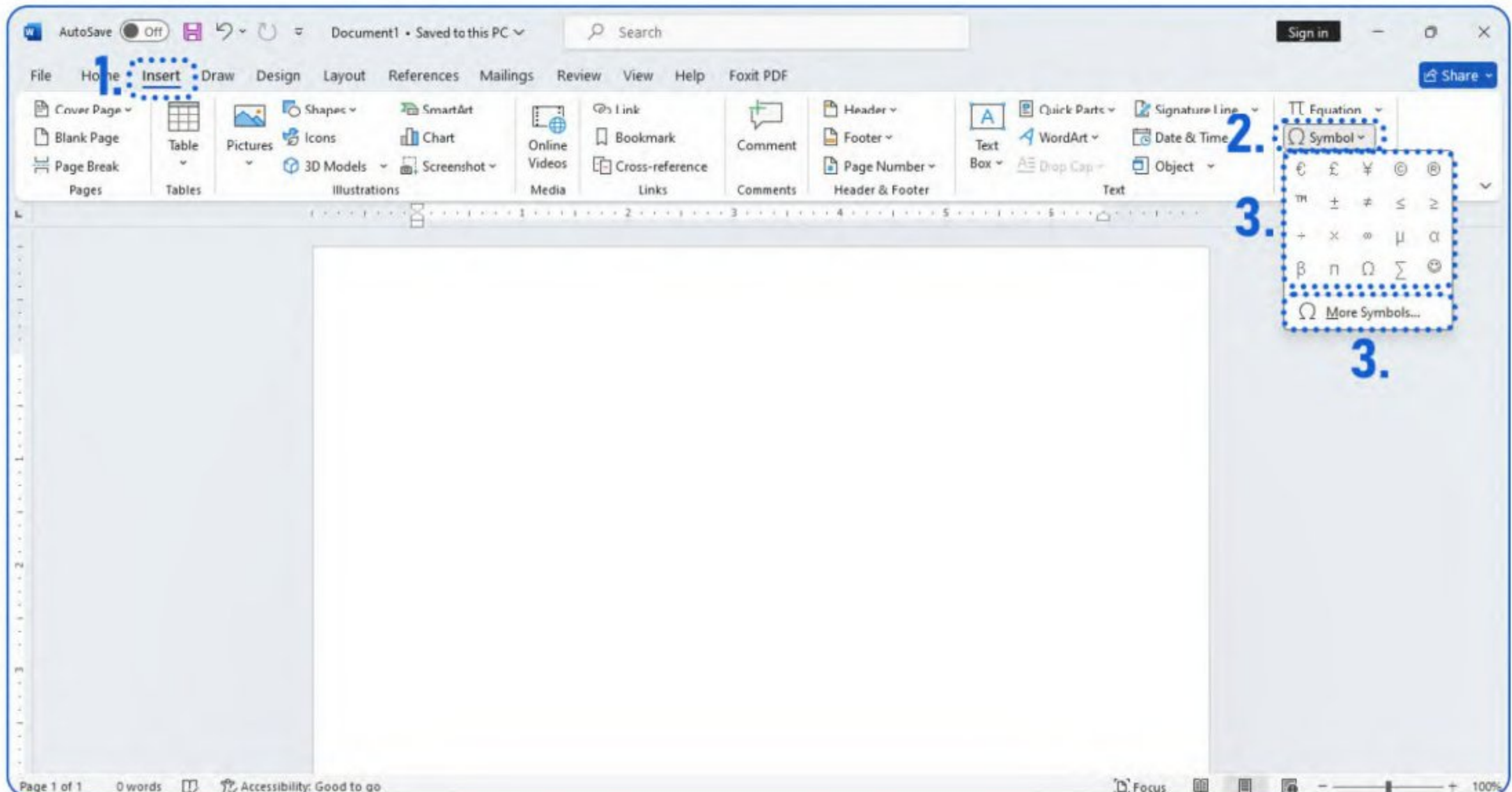
1. Click on the **Insert** tab.
2. In the **Header and Footer** section, you have all the options to apply and format header, footer and page numbers.



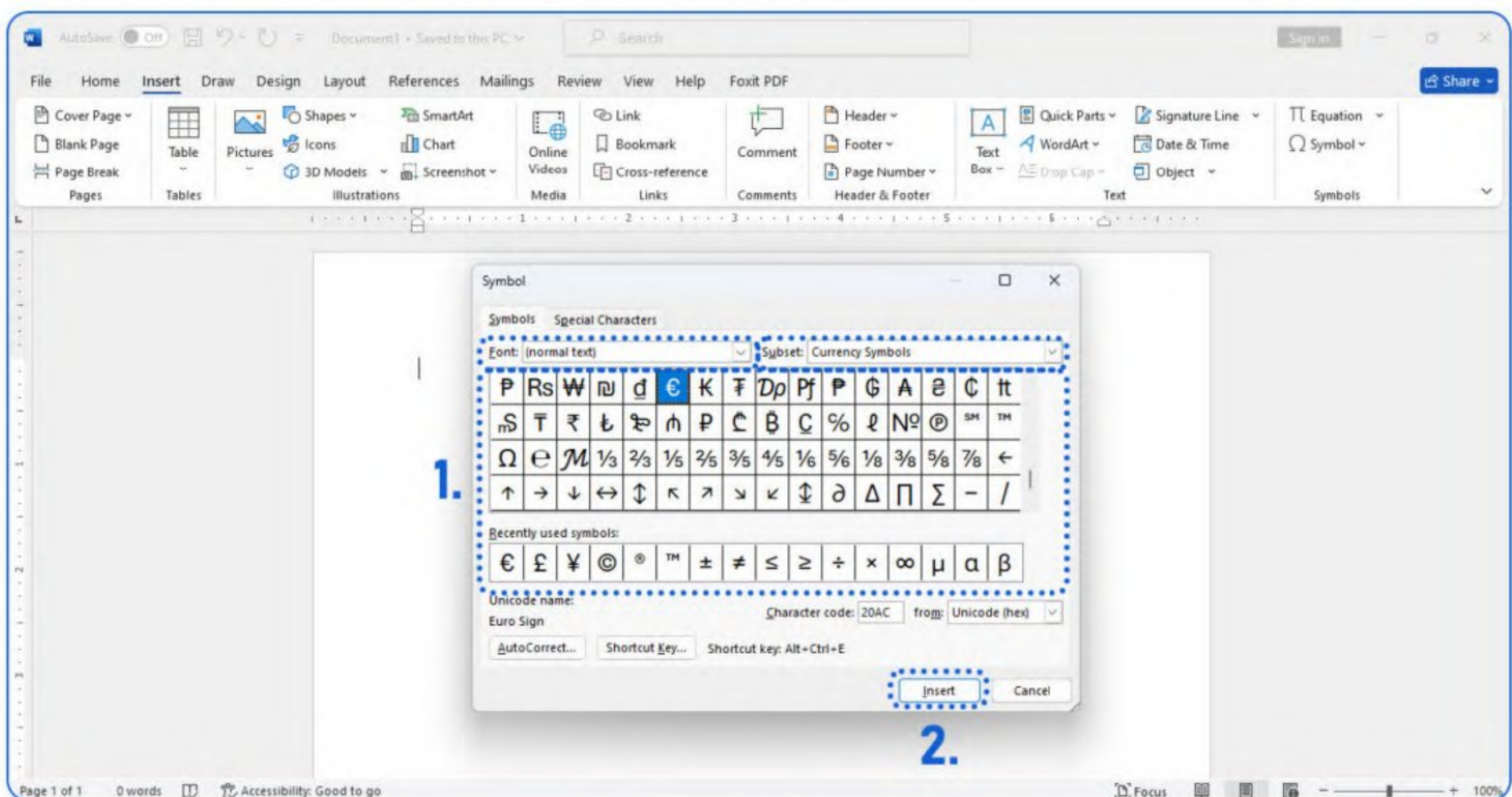


4. Inserting Symbols

1. Click on the **Insert** tab.
2. In the **Symbols** section, click on **Symbols**. A drop-down menu will appear.
3. You can select the most commonly used symbols suggested by **WORD** or you can click on **More Symbols...** to add other symbols.

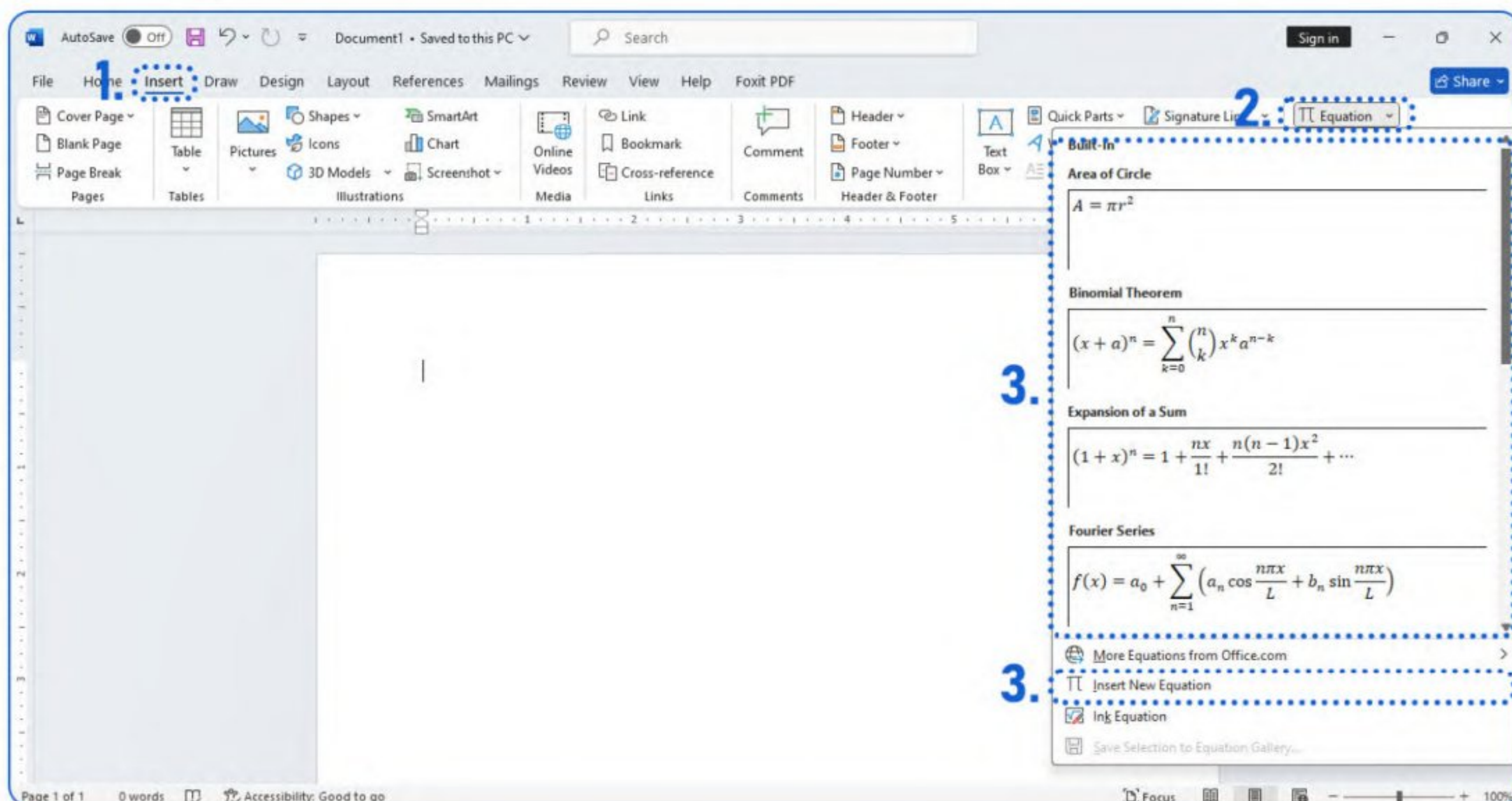


4. The **Symbol** window will appear where you can choose the **font** for the symbol, and the **type** of symbols.
5. Click on **Insert** after choosing the symbol.

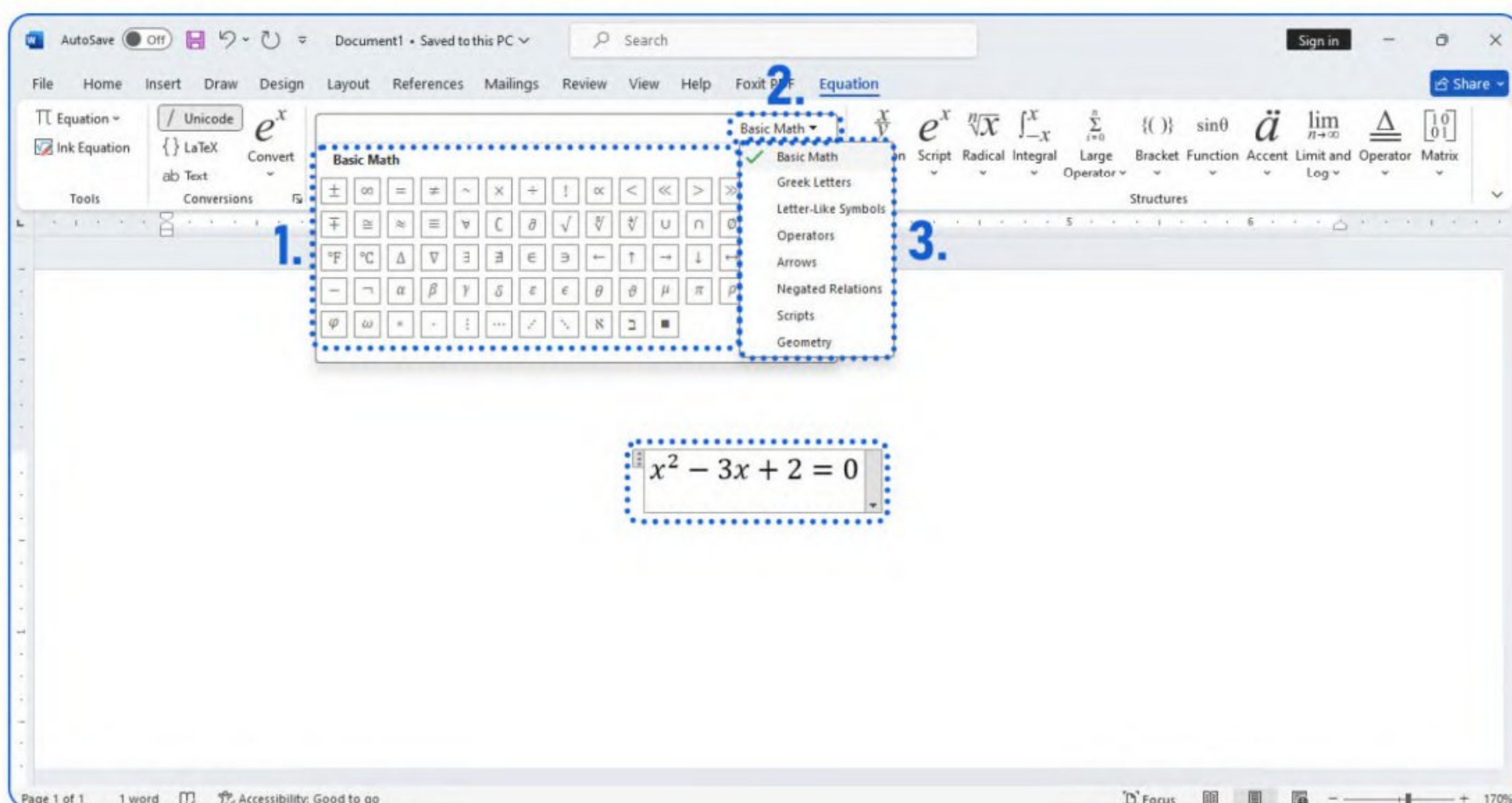


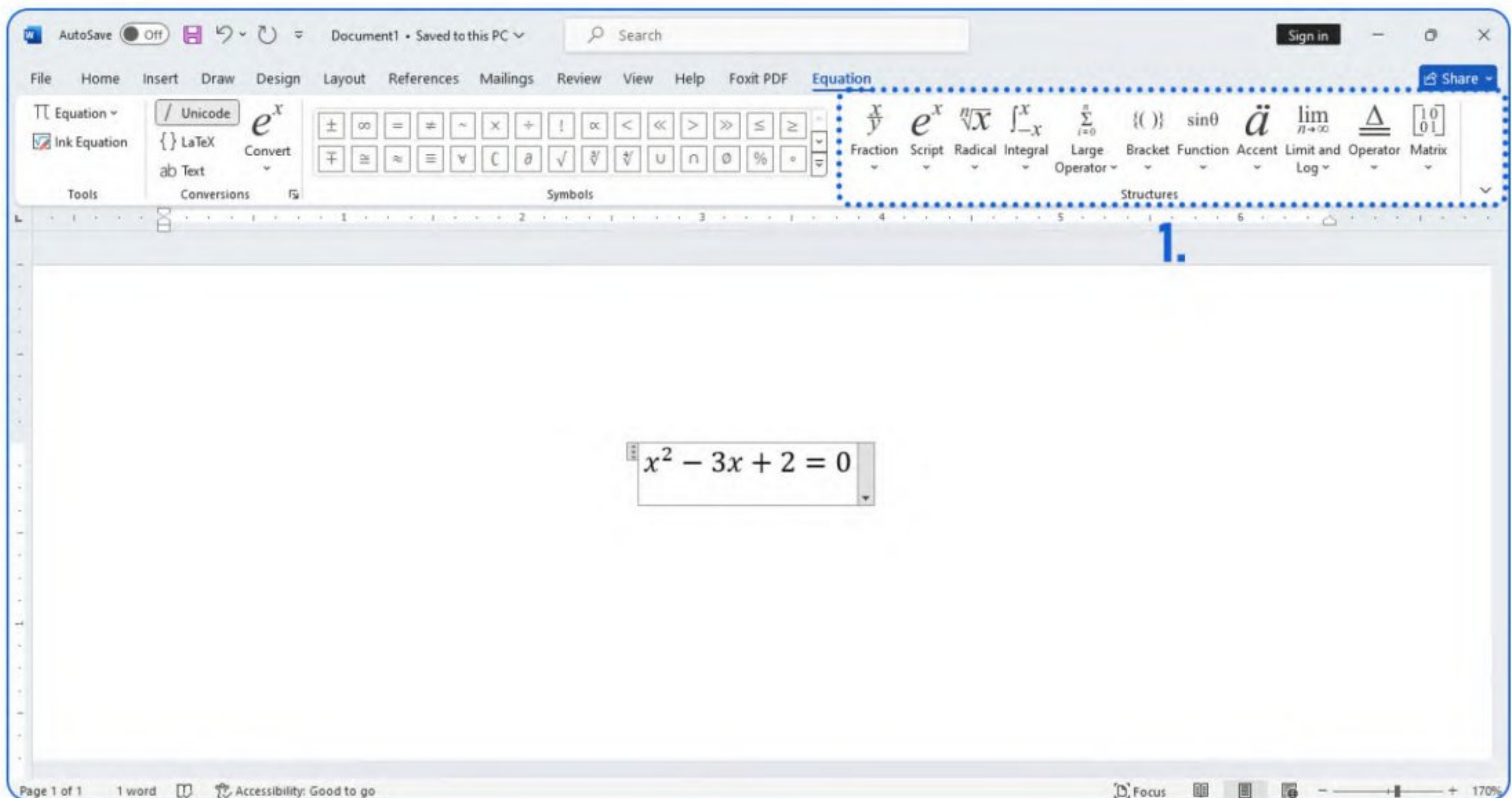
5. Inserting Equations

1. Click on the **Insert** tab.
2. In the **Symbols** section, click on **Equation**. A drop-down menu will appear.
3. You have a list of equations to choose. Or you can click on **Insert New Equation** to type the equation manually.



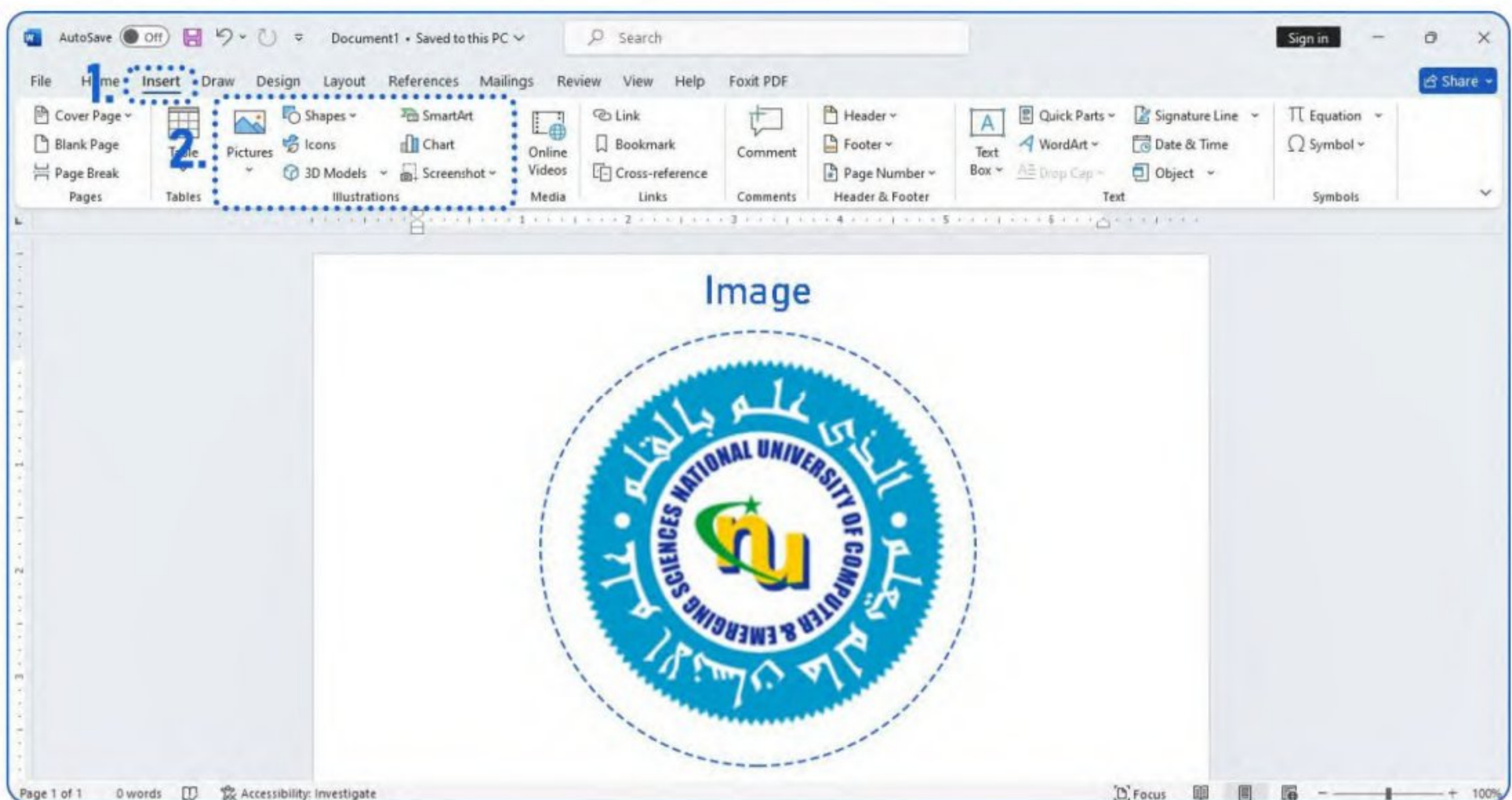
4. Now you can type the equation in the equation editor.
5. You can insert symbols and pre-designed equations structures under the **Equation** tab from the **Symbols** and **Structures** sections.





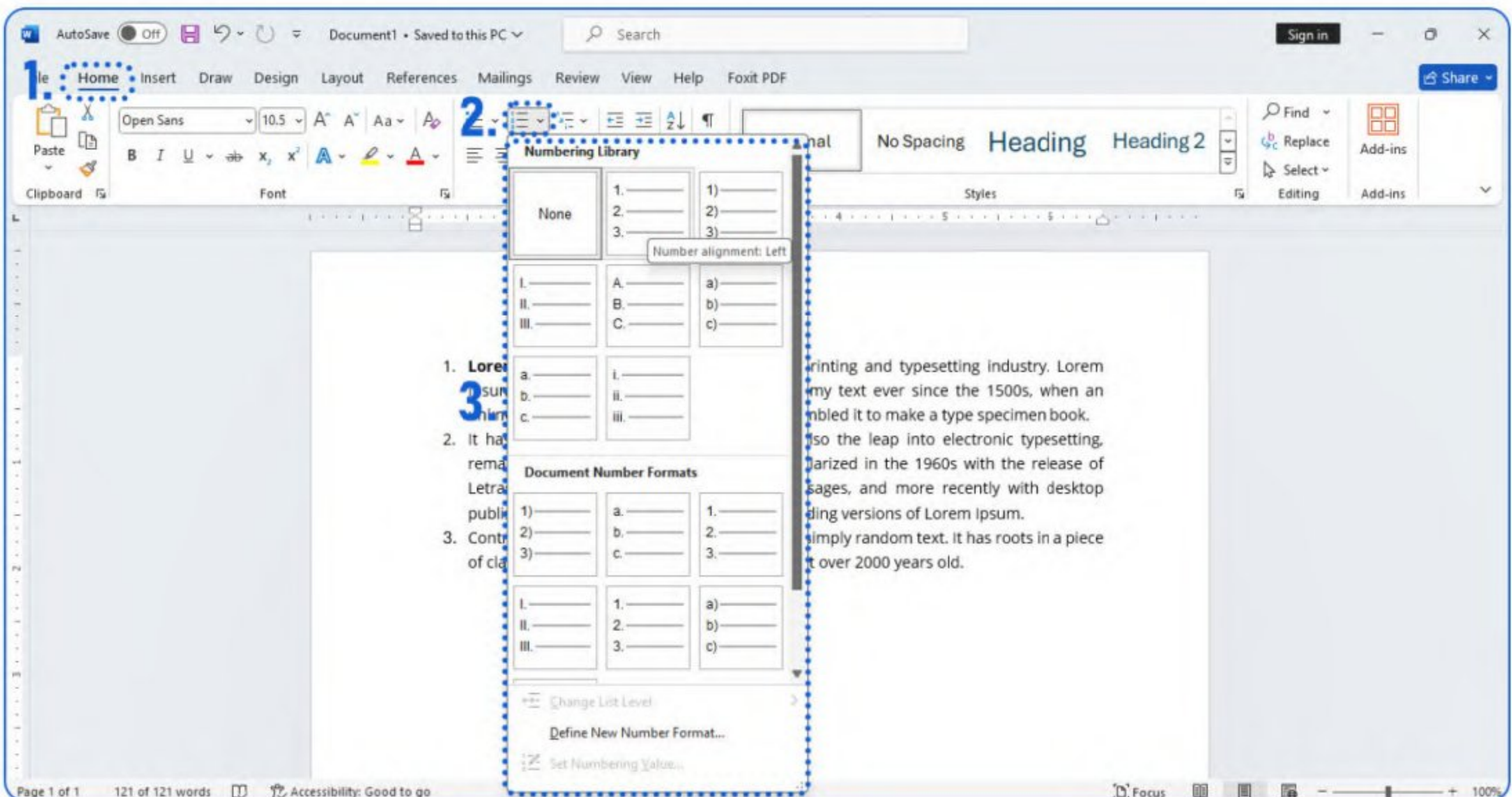
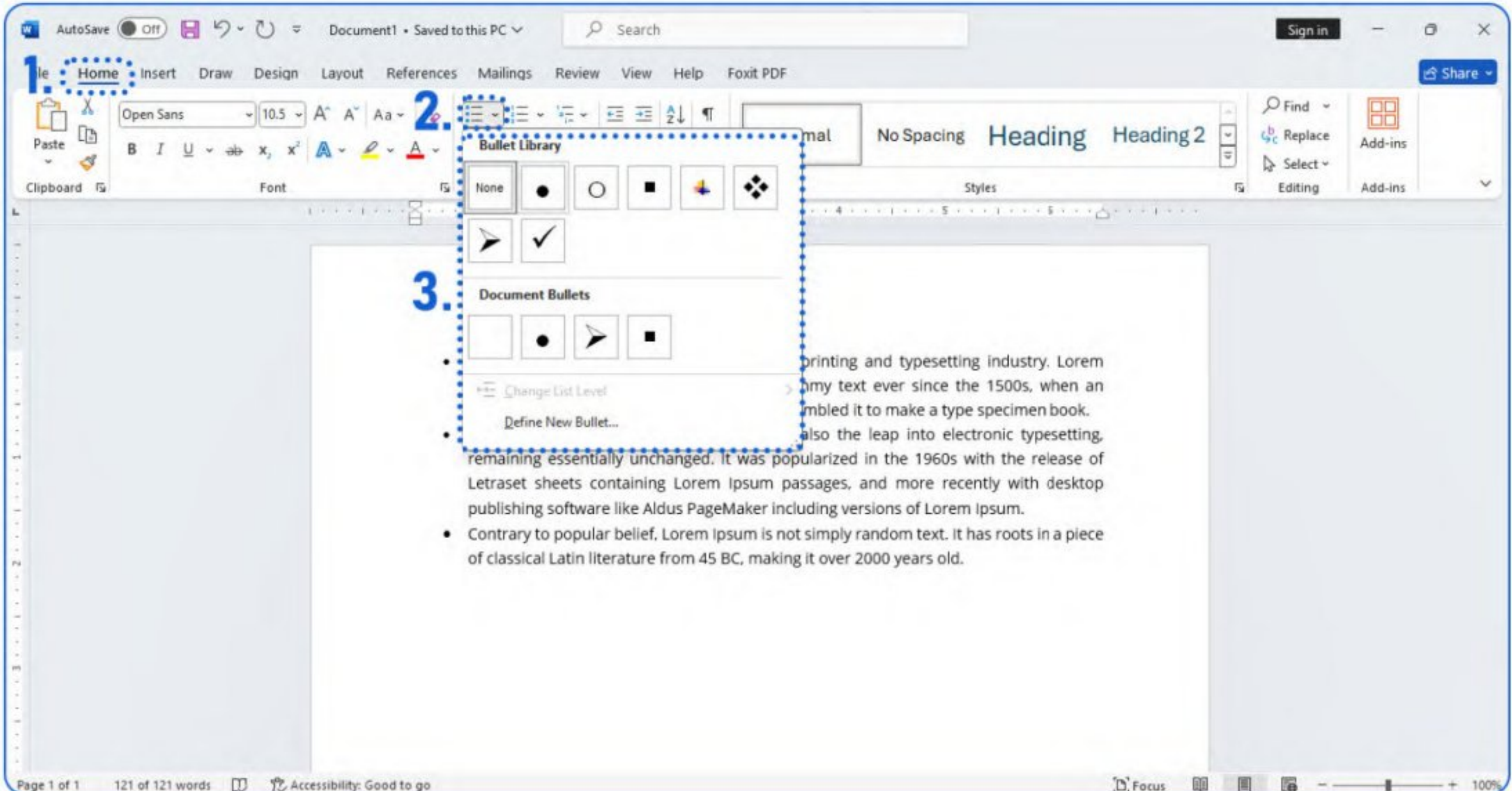
6. Inserting Images and Shapes

1. Click on the **Insert** Tab.
2. In the **Illustration** section, you have different options like inserting an image, inserting shaped and other options.



7. Inserting Bullets and Numbering

1. Select your text on which you want to apply bullets and numbering.
2. Click on the **Home** tab.
3. In the **Paragraph** section, click on the **bullet** icon to apply bullets on the select text.
4. Similarly, click on the **numbering** icon to apply number on your text.
5. You can also create intended bullets and numbering.



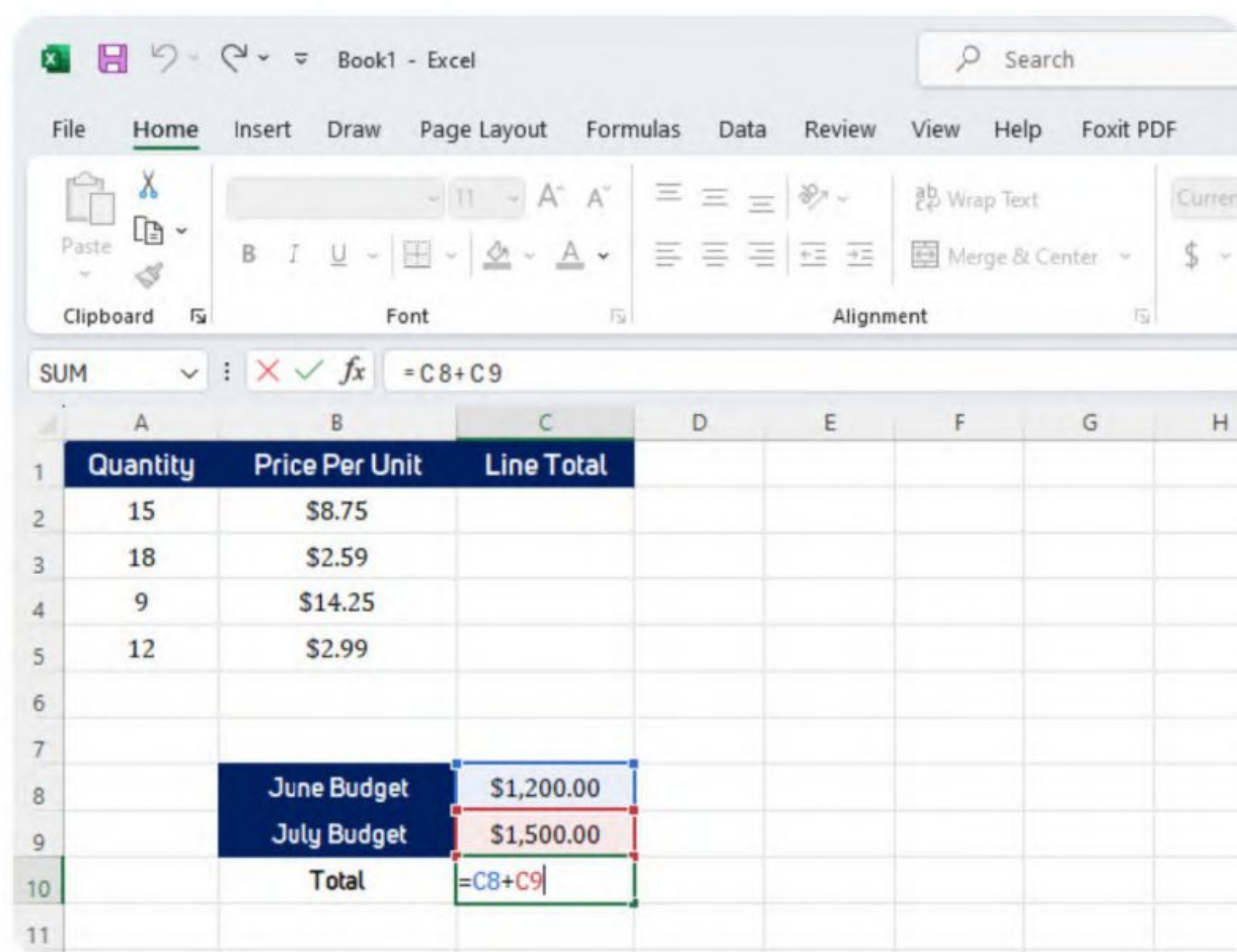
8. Introduction to Formulas in MS Excel

Excel uses standard operators for formulas: a plus sign for addition (+), minus sign for subtraction (-), asterisk for multiplication (*), forward slash for division (/), and caret (^) for exponents

Operation	Operator/Symbol	Formula	Description
Addition	+	=A1 + A2	Adds cells A1 & A2
Subtraction	—	=C4-3	Subtracts 3 from cell C4
Multiplication	*	E7/J4	Divides cell E7 by J4
Division	/	=N10*1.05	Multiplies cell N10 by 1.05
Exponents	^	R5^2	Finds the square of cell R5

9. Create Formula

1. Select the cell that will contain the formula. In our example, we'll select cell **C10**.
2. Type the equals sign (=). Select cell addresses you want to add. Type the mathematical operator you want to use. In our example, we'll type the addition sign (+).



	A	B	C	D	E	F	G	H
1	Quantity	Price Per Unit	Line Total					
2	15	\$8.75						
3	18	\$2.59						
4	9	\$14.25						
5	12	\$2.99						
6								
7								
8		June Budget	\$1,200.00					
9		July Budget	\$1,500.00					
10		Total	\$2,700.00					
11								

10. Functions

In order to work correctly, a function must be written a specific way, which is called the syntax. The basic syntax for a function is the equals sign (=), the function name (SUM, for example), and one or more arguments. Arguments contain the information you want to calculate. The function in the example below would add the values of the cell range **A1:A20**.

10.1 Creating a Function

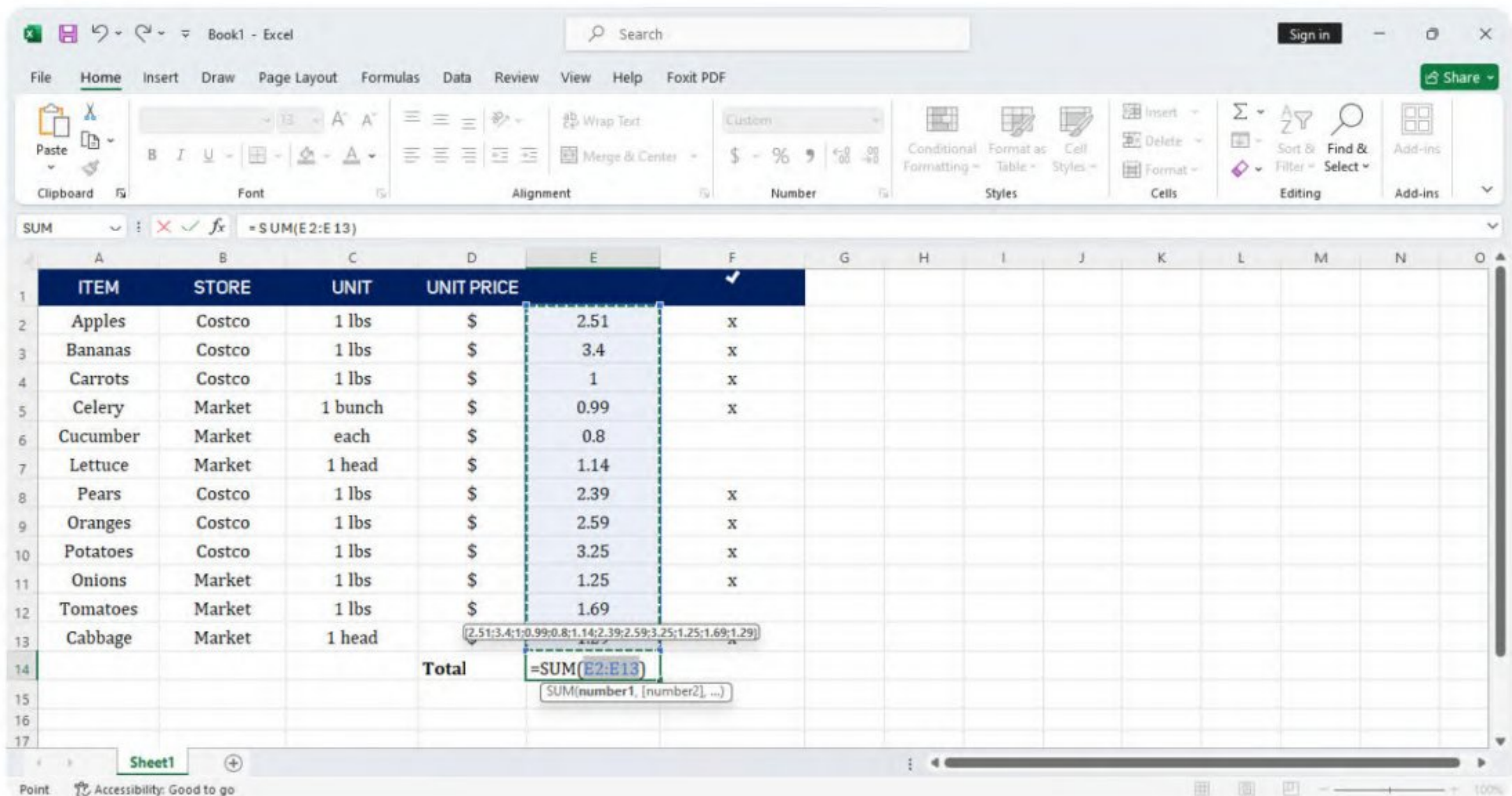
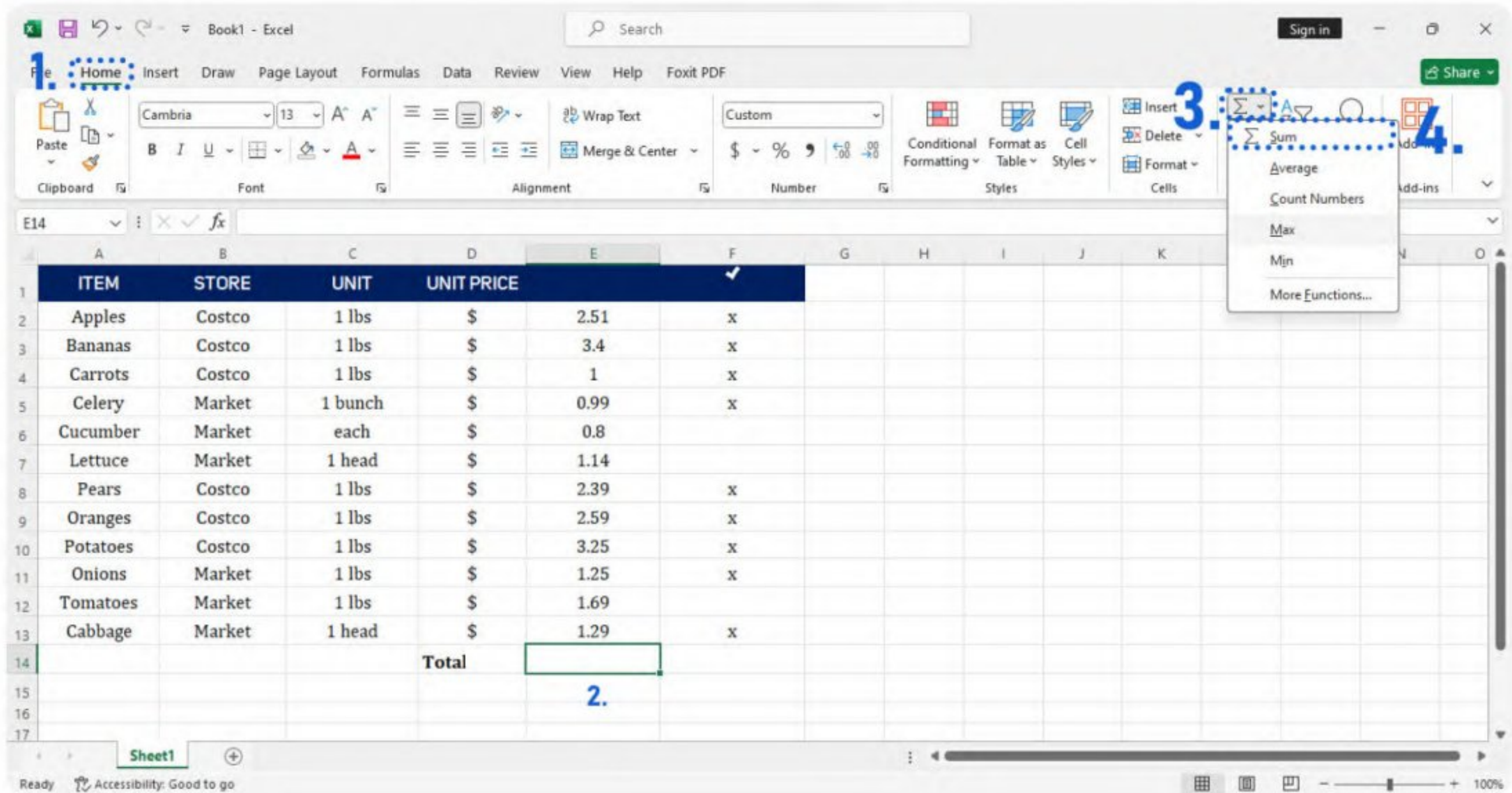
There are a variety of functions available in Excel. Here are some of the most common functions you'll use:

- SUM:** This function adds all of the values of the cells in the argument.
- AVERAGE:** This function determines the average of the values included in the argument. It calculates the sum of the cells and then divides that value by the number of cells in the argument.
- COUNT:** This function counts the number of cells with numerical data in the argument. This function is useful for quickly counting items in a cell range.
- MAX:** This function determines the highest cell value included in the argument.
- MIN:** This function determines the lowest cell value included in the argument.

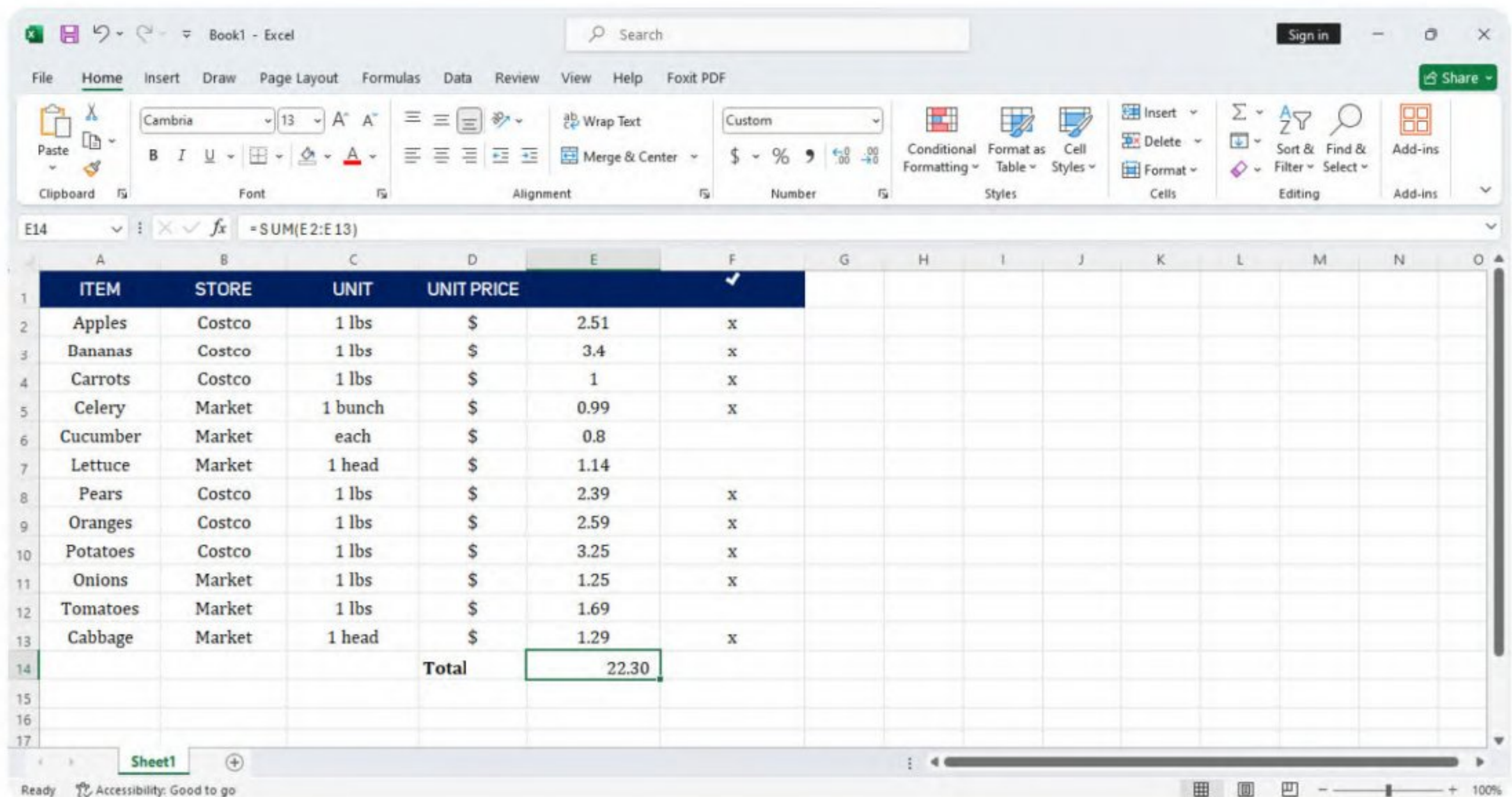
10.1 Creating a Function using the AutoSum Option

There are a variety of functions available in Excel. Here are some of the most common functions you'll use:

1. Click on the **Home** tab.
2. Select the cell that is below the numeric values you want to sum.
3. In the **Editing** group under the **Home** tab, click on the **summation** icon and then click on **Sum**.



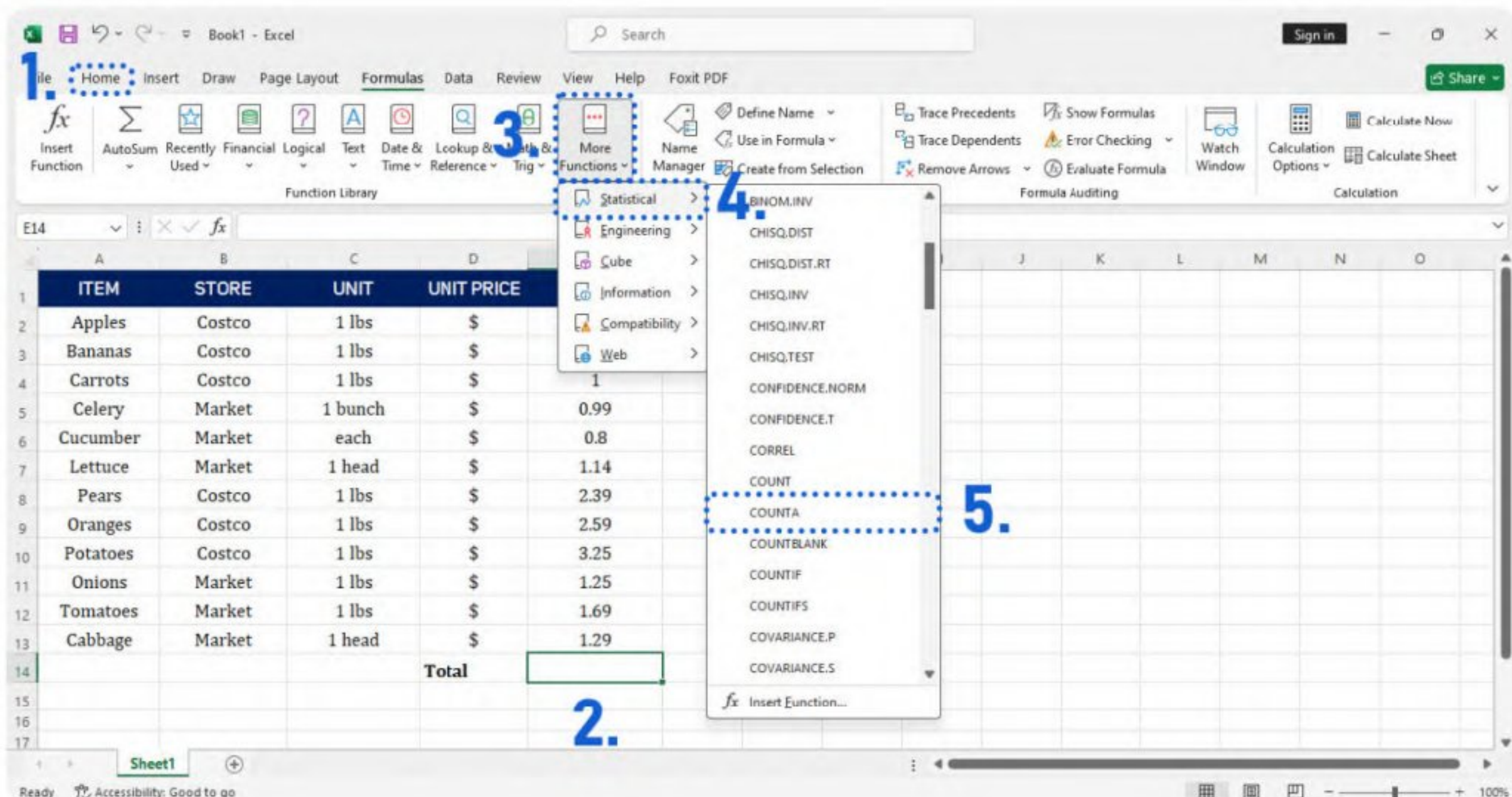
4. Press Enter.



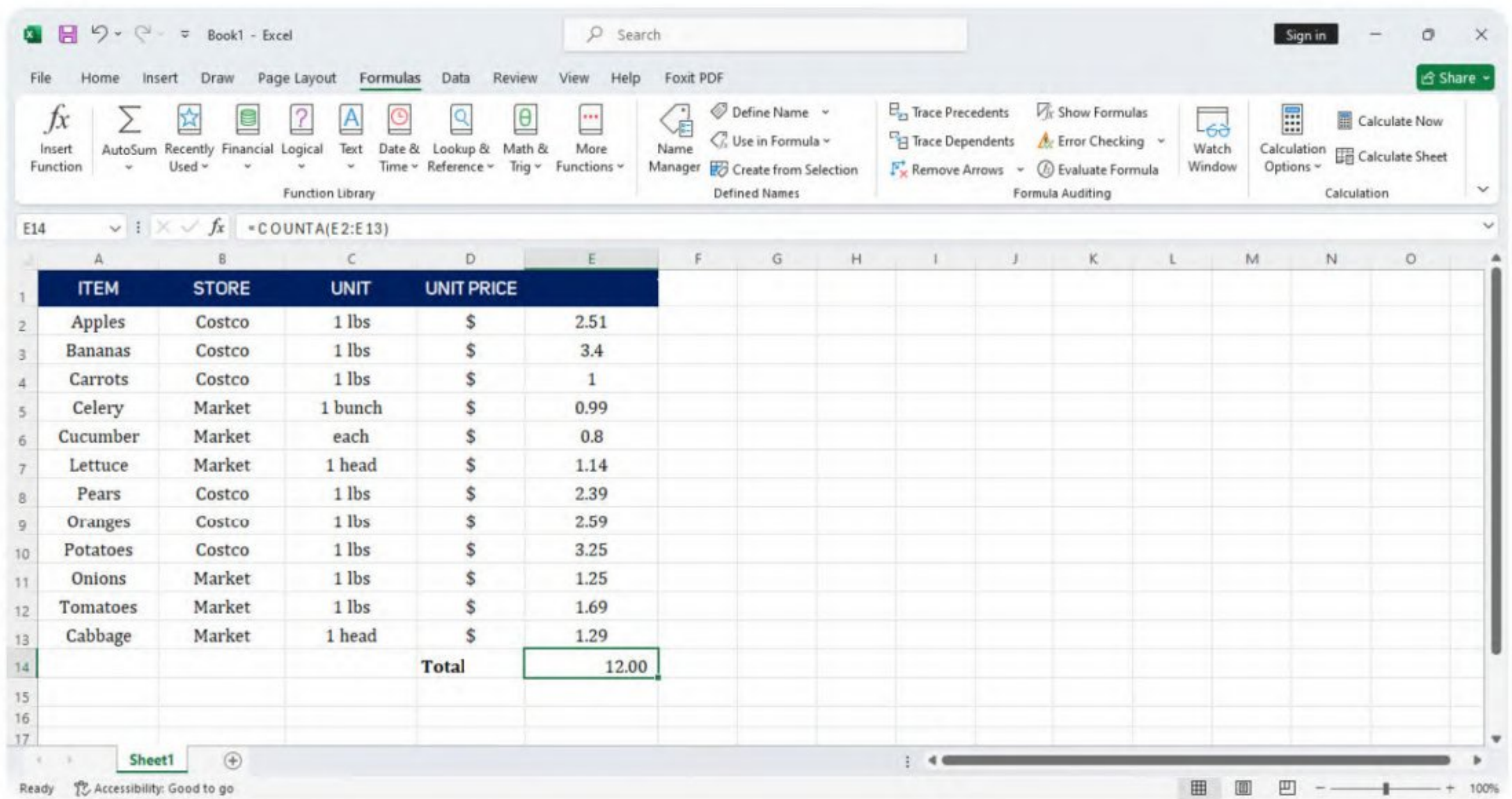
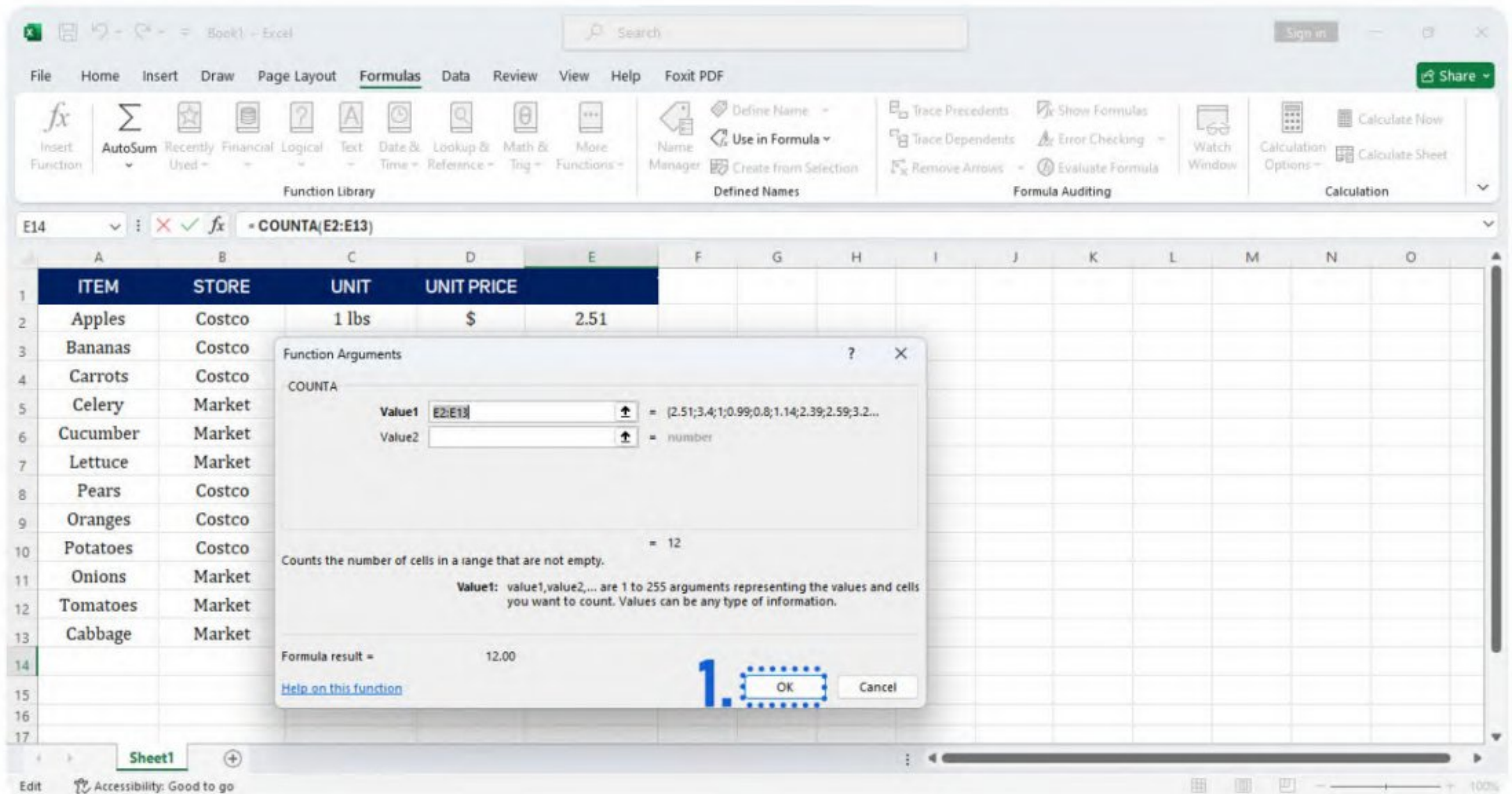
Note: The AutoSum option can also be accessed under the Formulas tab.

10.2 Inserting a Function from the Function Library

1. Click on the **Home** tab.
2. Click on the cell you want to apply the function. In our case it is **E14**.
3. In the **Function Library** section, click on **More Functions**. A drop-down menu will appear.
4. Click on **Statistical**. Now click on **COUNTA**. (This function will count the non-empty cells.)



5. The **Function Arguments** window will appear. The range will automatically be selected.
6. Click on **OK**.



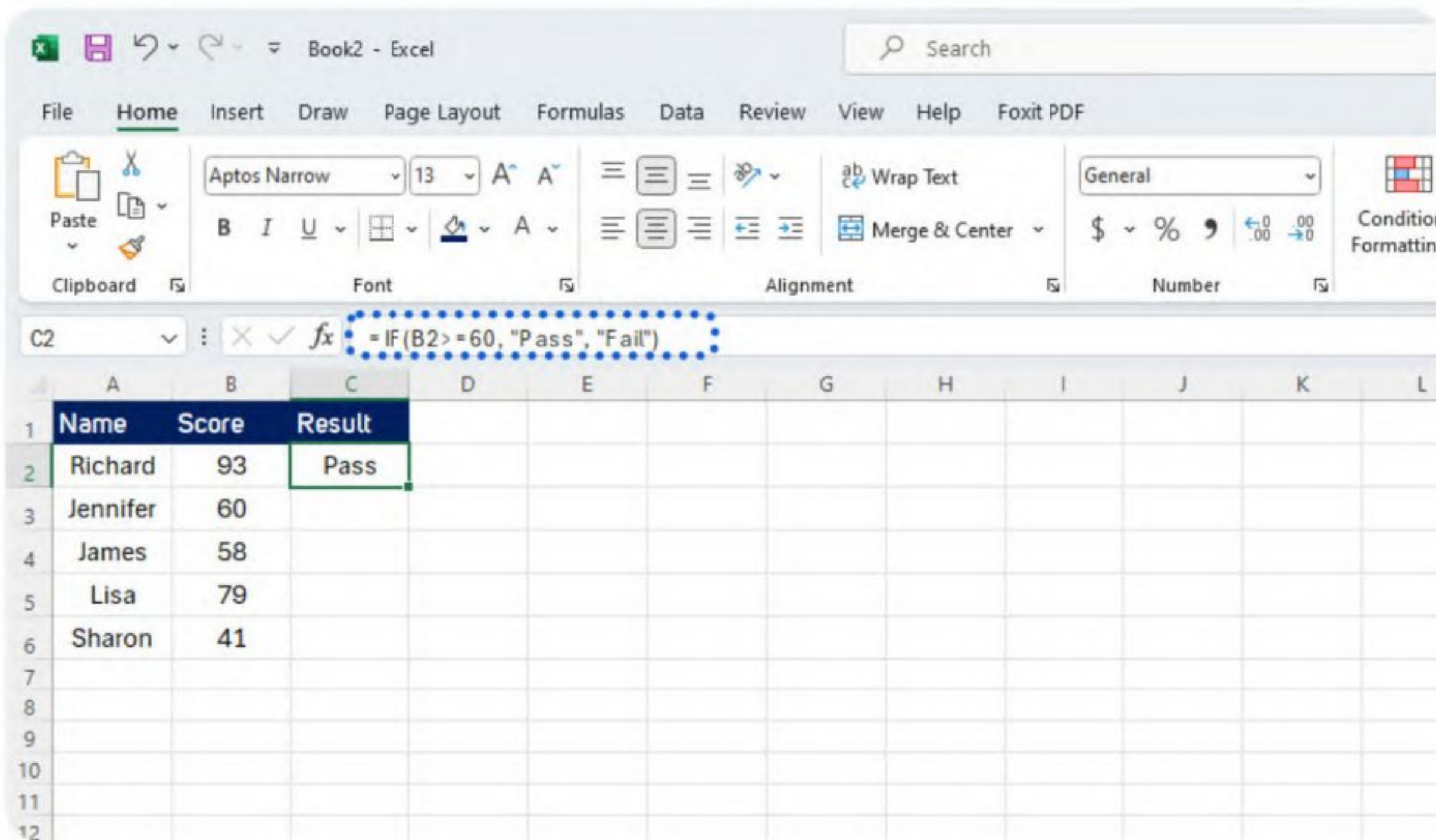
Hence, the items added to the cart are **12**.

11. Logical Functions

11.1 IF

The **IF** function checks whether a condition is met, and returns one value if true and another value if false.

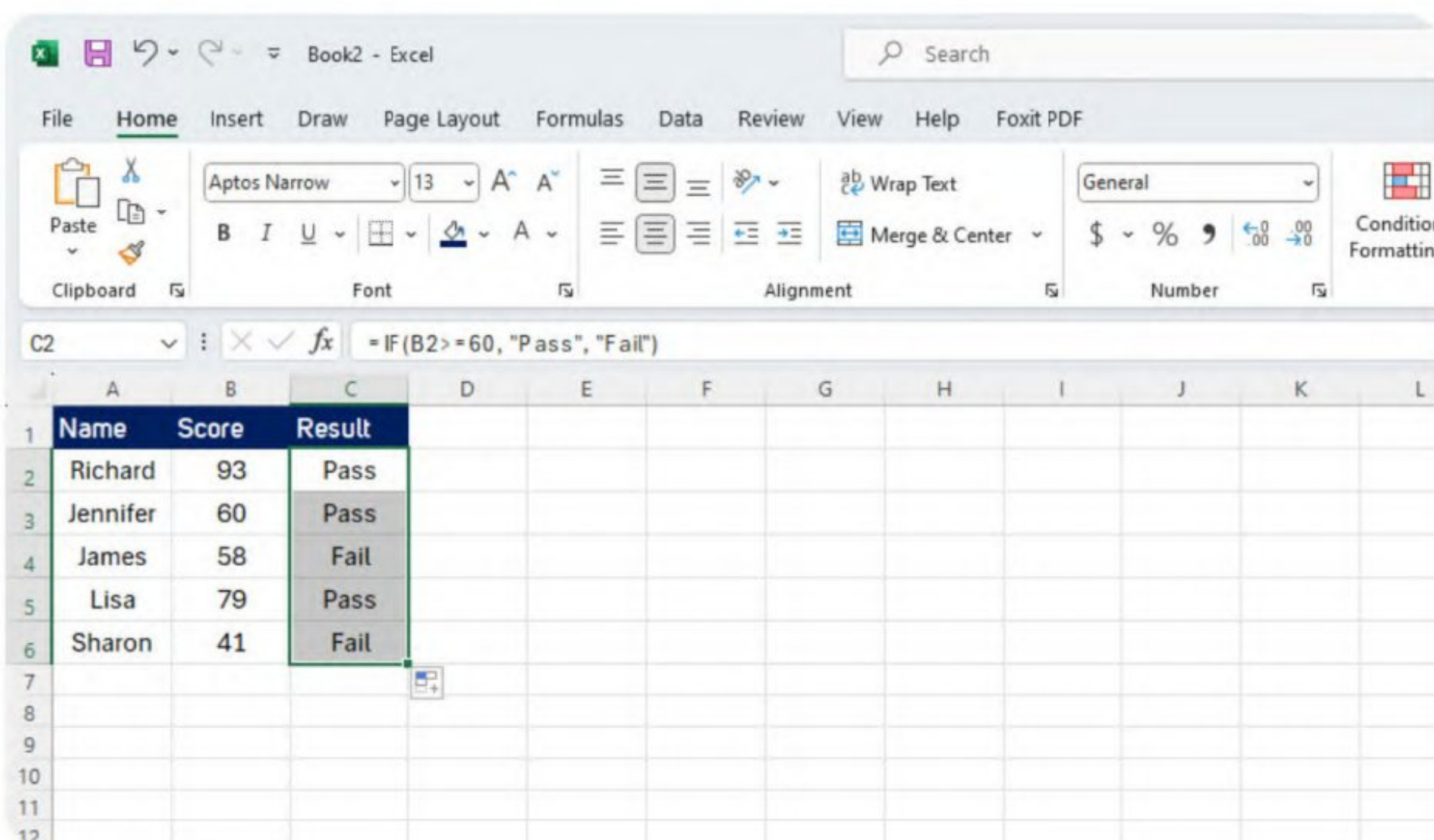
For example, take a look at the **IF** function in cell **C2** below:



The screenshot shows the Microsoft Excel interface with the 'Home' tab selected. The formula bar at the top displays the formula `=IF(B2>=60,"Pass","Fail")` for cell C2. The spreadsheet contains the following data:

	A	B	C	D	E	F	G	H	I	J	K	L
1	Name	Score	Result									
2	Richard	93	Pass									
3	Jennifer	60										
4	James	58										
5	Lisa	79										
6	Sharon	41										
7												
8												
9												
10												
11												
12												

Now, drag the handle (the **+** sign at the bottom-right corner of the selected cell) to all the cells to apply the same formula to other values of **Score**.

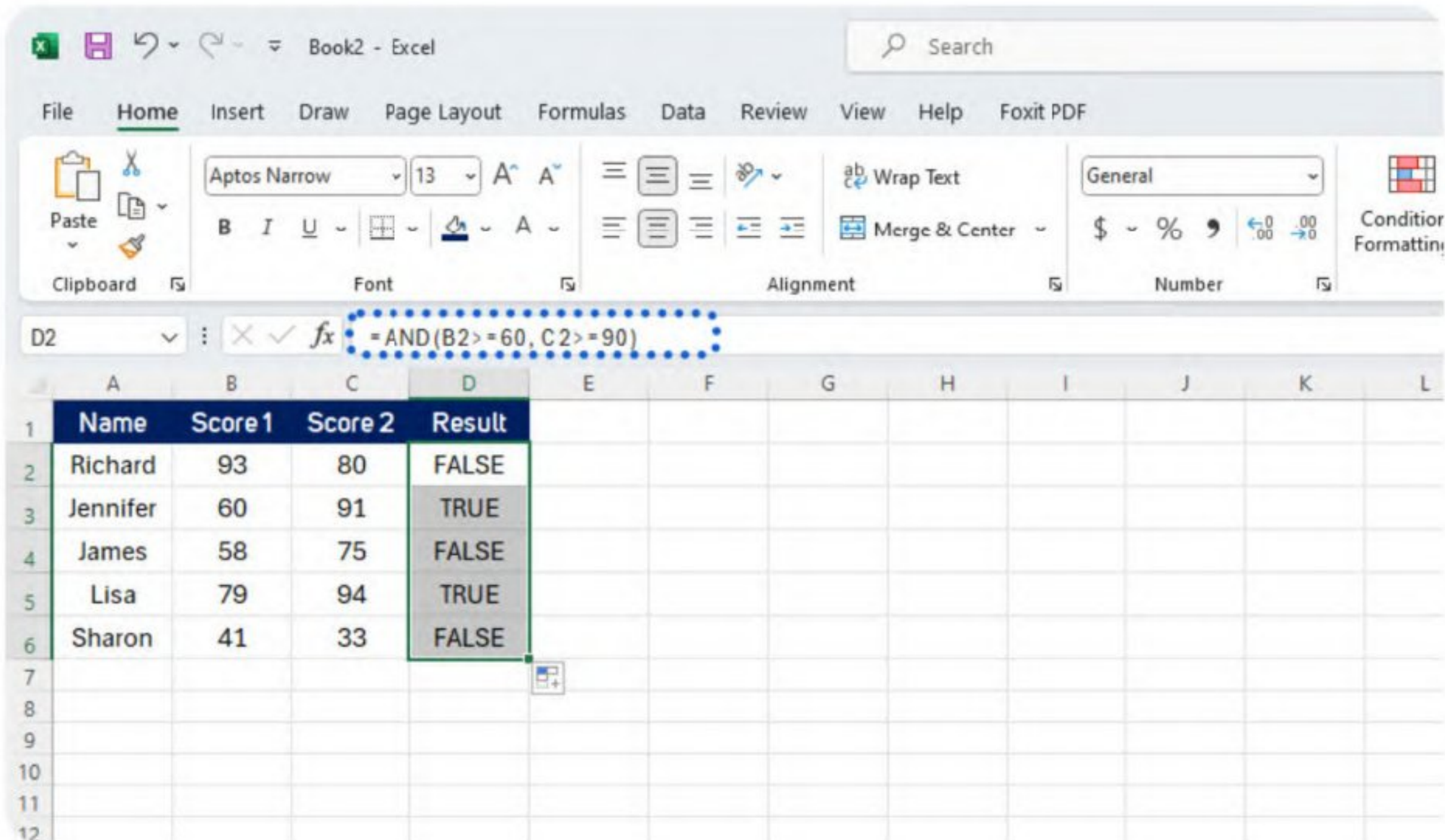


The screenshot shows the same Excel spreadsheet as before, but now the formula from cell C2 has been dragged down to cells C3 through C6. The results are as follows:

	A	B	C	D	E	F	G	H	I	J	K	L
1	Name	Score	Result									
2	Richard	93	Pass									
3	Jennifer	60	Pass									
4	James	58	Fail									
5	Lisa	79	Pass									
6	Sharon	41	Fail									
7												
8												
9												
10												
11												
12												

11.2 AND

The **AND** Function returns **TRUE** if all conditions are true and returns **FALSE** if any of the conditions are false.



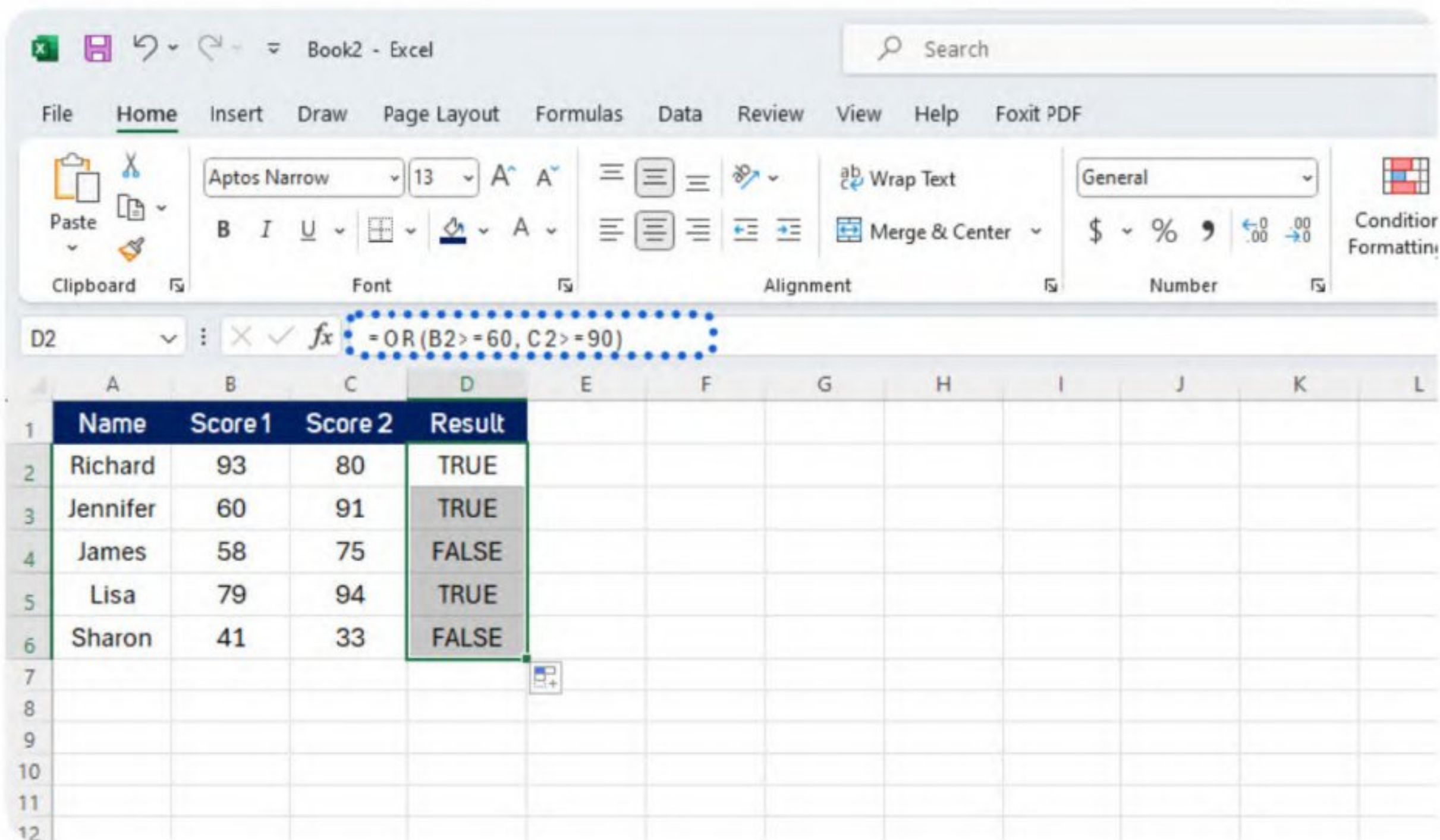
The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K	L
1	Name	Score 1	Score 2	Result								
2	Richard	93	80	FALSE								
3	Jennifer	60	91	TRUE								
4	James	58	75	FALSE								
5	Lisa	79	94	TRUE								
6	Sharon	41	33	FALSE								
7												
8												
9												
10												
11												
12												

The formula bar shows the formula in cell D2: `=AND(B2>=60, C2>=90)`.

11.3 OR

The **OR** function returns **TRUE** if any of the conditions are **TRUE** and returns **FALSE** if all conditions are false.



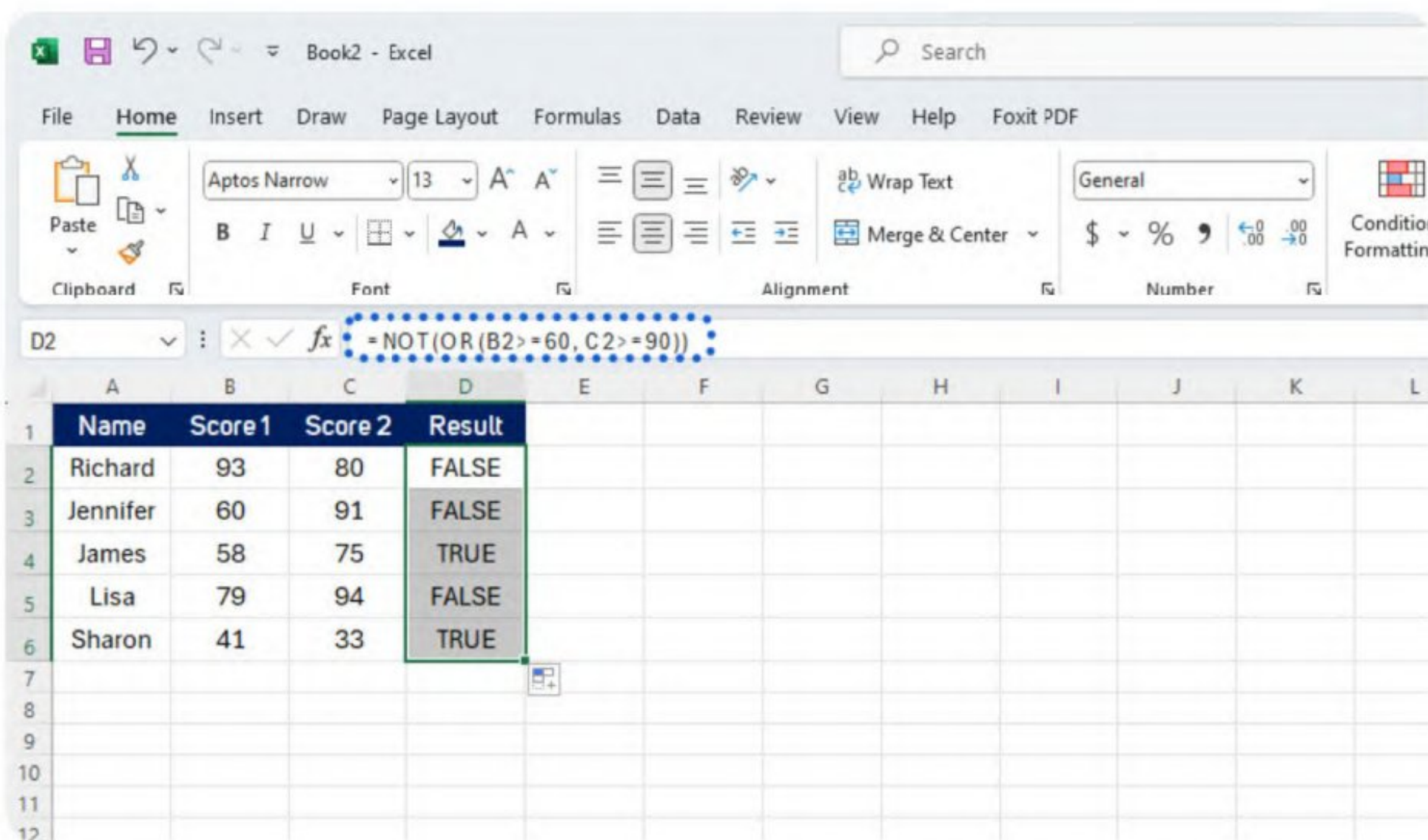
The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K	L
1	Name	Score 1	Score 2	Result								
2	Richard	93	80	TRUE								
3	Jennifer	60	91	TRUE								
4	James	58	75	FALSE								
5	Lisa	79	94	TRUE								
6	Sharon	41	33	FALSE								
7												
8												
9												
10												
11												
12												

The formula bar shows the formula in cell D2: `=OR(B2>=60, C2>=90)`.

11.4 NOT

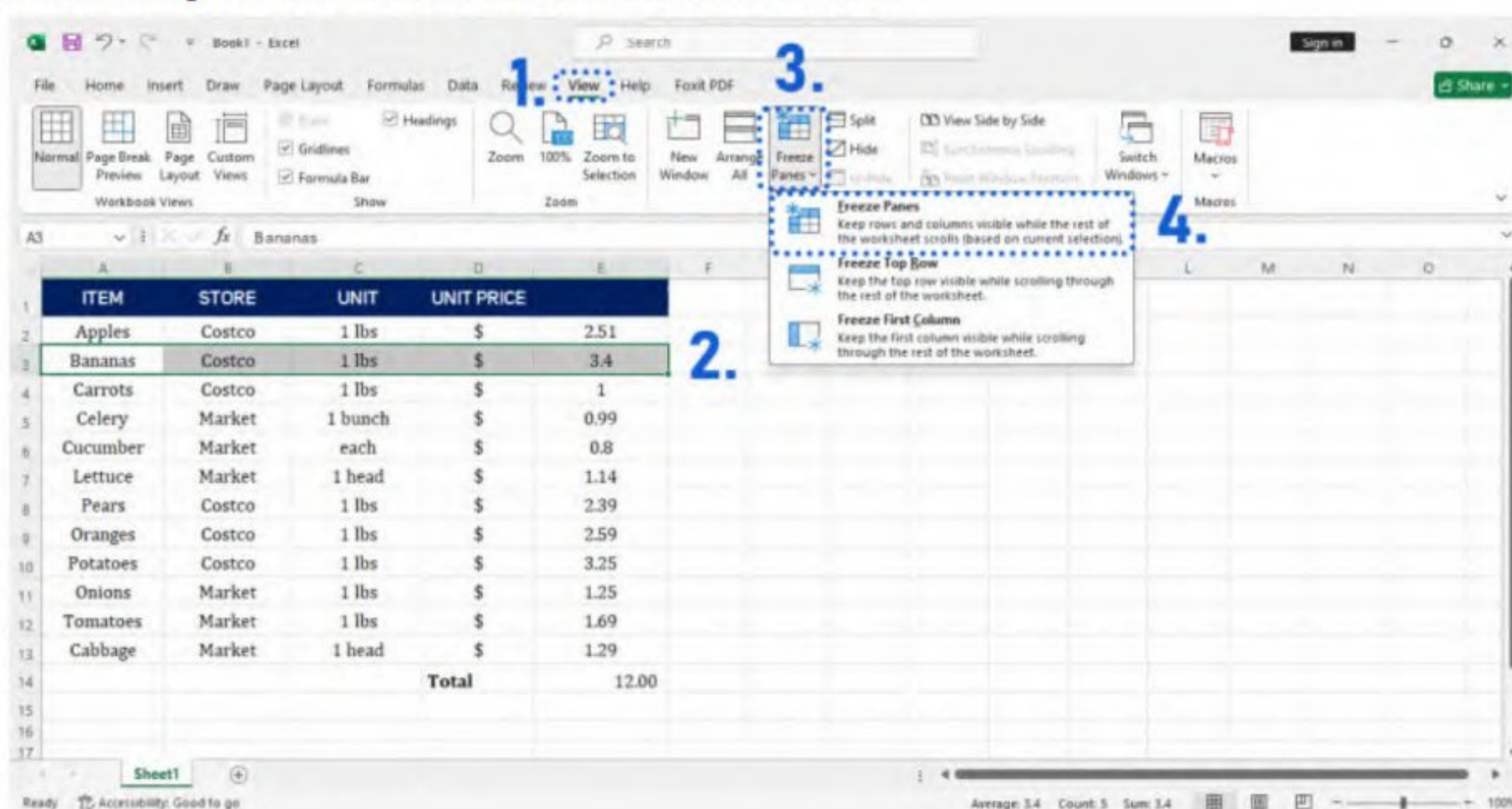
The **NOT** function changes **TRUE** to **FALSE**, and **FALSE** to **TRUE**.



12. Freeze Rows

You may want to see certain rows or columns all the time in your worksheet, especially header cells. By freezing rows or columns in place, you'll be able to scroll through your content while continuing to view the frozen cells.

1. Click on the **View** tab.
2. To freeze the **first two rows**, (row 1 and 2) select the **3rd row** so that you can scroll all the rows except row 1 and 2.
3. Under the **Window** section, click on **Freeze Panes**.
4. In the drop-down menu, click on **Freeze Panes**.



ITEM	STORE	UNIT	UNIT PRICE
Apples	Costco	1 lbs	\$ 2.51
Lettuce	Market	1 head	\$ 1.14
Pears	Costco	1 lbs	\$ 2.39
Oranges	Costco	1 lbs	\$ 2.59
Potatoes	Costco	1 lbs	\$ 3.25
Onions	Market	1 lbs	\$ 1.25
Tomatoes	Market	1 lbs	\$ 1.69
Cabbage	Market	1 head	\$ 1.29
Total			12.00

Note: Do the same to freeze the Columns.

Note: To unfreeze panes, simply click on Unfreeze Panes.

13. Sorting a Range

1. Click on the **Home** tab.
2. Select the column to sort.
3. In the **Editing** section, click on **Sort & Filter**.
4. You can sort from **Smallest to Largest**, **Largest to Smallest**, and **Custom Sort** based on your sorting criteria.
5. Select **Largest to Smallest** option to sort the data based on the **Quantity** column.

ITEM	STORE	UNIT	QUANTITY	UNIT PRICE
Apples	Costco	1 lbs	7	\$ 2.51
Bananas	Costco	1 lbs	5	\$ 3.4
Carrots	Costco	1 lbs	2	\$ 1
Celery	Market	1 bunch	9	\$ 0.99
Cucumber	Market	each	4	\$ 0.8
Lettuce	Market	1 head	6	\$ 1.14
Pears	Costco	1 lbs	7	\$ 2.39
Oranges	Costco	1 lbs	12	\$ 2.59
Potatoes	Costco	1 lbs	6	\$ 3.25
Onions	Market	1 lbs	3	\$ 1.25
Tomatoes	Market	1 lbs	4	\$ 1.69
Cabbage	Market	1 head	9	\$ 1.29
Total				12.00

6. After this, the **Sort Warning** window will appear, click on **OK**.

Link: [Grocery Data](#)

Sort Warning

Microsoft Excel found data next to your selection. Since you have not selected this data, it will not be sorted.

What do you want to do?

☒ Expand the selection

☐ Continue with the current selection

1. Sort Cancel

ITEM	STORE	UNIT	QUANTITY	UNIT PRICE
Apples	Costco	1 lbs	7	\$ 2.51
Bananas	Costco	1 lbs	5	\$ 3.25
Carrots	Costco	1 lbs	2	\$ 1.25
Celery	Market	1 bunch	9	\$ 1.69
Cucumber	Market	each	4	\$ 1.29
Lettuce	Market	1 head	6	\$ 1.29
Pears	Costco	1 lbs	7	\$ 1.29
Oranges	Costco	1 lbs	12	\$ 1.29
Potatoes	Costco	1 lbs	6	\$ 1.29
Onions	Market	1 lbs	3	\$ 1.29
Tomatoes	Market	1 lbs	4	\$ 1.29
Cabbage	Market	1 head	9	\$ 1.29
Total				12.00

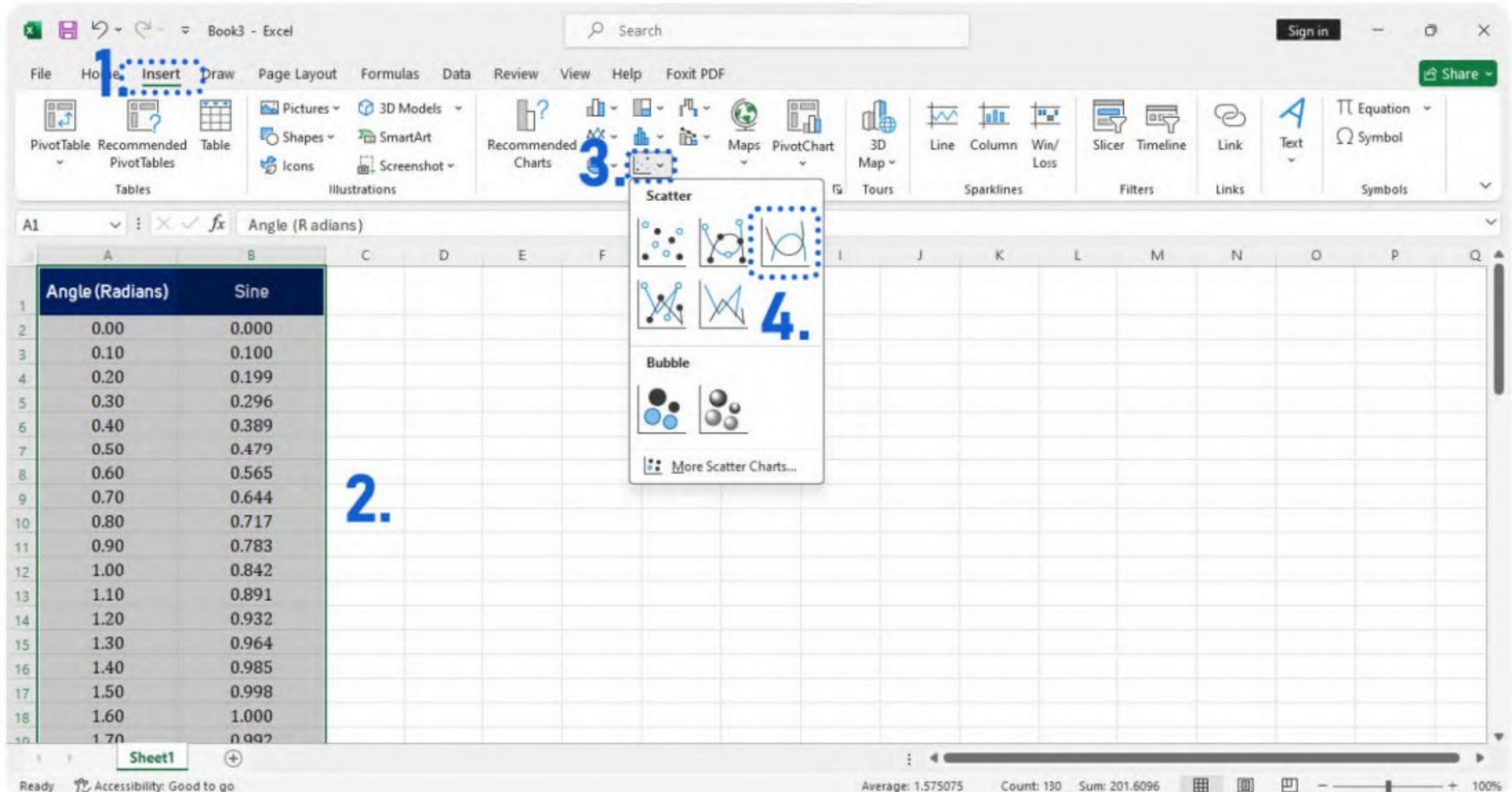
ITEM	STORE	UNIT	QUANTITY	UNIT PRICE
Oranges	Costco	1 lbs	12	\$ 2.59
Celery	Market	1 bunch	9	\$ 0.99
Cabbage	Market	1 head	9	\$ 1.29
Apples	Costco	1 lbs	7	\$ 2.51
Pears	Costco	1 lbs	7	\$ 2.39
Lettuce	Market	1 head	6	\$ 1.14
Potatoes	Costco	1 lbs	6	\$ 3.25
Bananas	Costco	1 lbs	5	\$ 3.4
Cucumber	Market	each	4	\$ 0.8
Tomatoes	Market	1 lbs	4	\$ 1.69
Onions	Market	1 lbs	3	\$ 1.25
Carrots	Costco	1 lbs	2	\$ 1
Total				12.00

14. Charts in Excel

1. Click on the **Insert** tab.
2. Select the range(s) of data for which you want to create the chart. In this example, the data to create chart of a **Sine Wave** is used.

Link: [Sine Wave Data](#)

3. In the **Charts** section, click on the **Scatter Chart** icon.
4. Select the 3rd option to create a smooth line chart.



5. You can customize the **chart style**, charts element, chart title and many more.

